#1 · Medical Radiation Practitioner Core

What is the difference between ionising and non-ionising radiation, and which kind is used in MRI?

Answer

Ionising radiation has enough energy per photon or particle to remove an electron from an atom. Non-ionising radiation does not. MRI uses non-ionising magnetic fields and radiofrequency pulses.

#2 · Medical Radiation Practitioner Core

Two photons have identical energy. One is called an x-ray and one a gamma ray. What separates them?

Answer

Only their origin. X-rays come from the electron cloud or from electrons slowing down, while gamma rays come from the nucleus.

#3 · Medical Radiation Practitioner Core

In an x-ray tube, roughly what share of the electron energy becomes x-rays and what share becomes heat?

Answer

About 1 per cent becomes x-rays and about 99 per cent becomes heat. This is why anodes must cope with heat and loads are limited.

#4 · Medical Radiation Practitioner Core

A radiographer raises mA but keeps kVp the same. What changes in the beam?

Answer

Only the number of photons changes, not their energy. Raising kVp changes the photon energy, penetration and contrast.

#5 · Medical Radiation Practitioner Core

Which interaction gives most of the contrast in a diagnostic x-ray image, and how does it vary with Z and energy?

Answer

The photoelectric effect. It rises steeply with atomic number and falls steeply as photon energy rises, so bone and iodine absorb much more than soft tissue.

#6 · Medical Radiation Practitioner Core

Which interaction is the main source of scatter and staff dose in soft tissue?

Answer

Compton scatter. It depends on electron density rather than atomic number, and the patient becomes the source of scatter radiation.

#7 · Medical Radiation Practitioner Core

What is the minimum photon energy for pair production, and why does PET depend on a related process?

Answer

Pair production needs more than 1.022 MeV. PET depends on annihilation, where a positron and electron vanish and give two 511 keV photons in opposite directions.

#8 · Medical Radiation Practitioner Core

A beam of 100 µGy per hour passes through three half-value layers of lead. What intensity emerges?

Answer

100 × 0.5 × 0.5 × 0.5 = 12.5 µGy per hour.

#9 · Medical Radiation Practitioner Core

The air kerma rate is 4 mGy per hour at 1 m from a mobile unit. What is it at 2 m?

Answer

Doubling the distance divides the rate by four, so 4 ÷ 4 = 1 mGy per hour.

#10 · Medical Radiation Practitioner Core

A staff member must stay near a source during a bedside examination. Which protection action is most effective and cheapest?

Answer

Increase distance, because intensity falls with the square of distance. Reducing time and using shielding add further protection.

#11 · Medical Radiation Practitioner Core

A Tc-99m dose is 600 MBq at 08:00 with a half-life of 6 hours. What activity remains at 14:00?

Answer

14:00 is one half-life later, so the activity is 600 × 0.5 = 300 MBq.

#12 · Medical Radiation Practitioner Core

A nuclear medicine patient arrives two hours late for a dose drawn up at calibration time. What is the best action?

Answer

Recalculate the activity for the actual injection time using the half-life, then adjust the volume or draw a new dose, so the patient gets the prescribed activity.

#13 · Medical Radiation Practitioner Core

What is the formula that links absorbed dose, equivalent dose and effective dose?

Answer

Equivalent dose = absorbed dose × wR. Effective dose = sum over tissues of equivalent dose × wT, and the wT values add up to 1.

#14 · Medical Radiation Practitioner Core

Lung receives 10 mGy and thyroid 5 mGy from photons. Using wT of 0.12 and 0.04, what is the effective dose from these two tissues?

Answer

wR is 1 for photons. Lung 10 × 0.12 = 1.2 mSv and thyroid 5 × 0.04 = 0.2 mSv, so the total is 1.4 mSv.

#15 · Medical Radiation Practitioner Core

What are the occupational and public effective dose limits in the standard ICRP and ARPANSA values?

Answer

Occupational: 20 mSv per year averaged over five years, with no more than 50 mSv in any single year. Public: 1 mSv per year.

#16 · Medical Radiation Practitioner Core

A pregnant patient is referred for a pelvic radiograph. What should the practitioner do first?

Answer

Tell the referring practitioner or radiologist, who decides whether the examination is justified, delayed or replaced by a non-ionising method. The practitioner should not cancel or proceed alone.

#17 · Medical Radiation Practitioner Core

What are the three principles of radiation protection used in medical radiation practice?

Answer

Justification, limitation and optimisation. Justification asks whether more good than harm results, limitation sets dose limits for workers and the public, and optimisation keeps doses as low as reasonably achievable.

#18 · Medical Radiation Practitioner Core

Do occupational dose limits apply to a patient having a justified X-ray examination?

Answer

No. Dose limits apply to workers and members of the public. A patient's dose is controlled by justification and optimisation.

#19 · Medical Radiation Practitioner Core

State the occupational effective dose limit used in Australia.

Answer

20 mSv per year averaged over five consecutive years, with no more than 50 mSv in any single year.

#20 · Medical Radiation Practitioner Core

What is the effective dose limit for a member of the public?

Answer

1 mSv per year.

#21 · Medical Radiation Practitioner Core

A worker's doses over five years are 18, 22, 15, 20 and 25 mSv. Is the occupational limit met?

Answer

Yes. The total is 100 mSv, the average is 20 mSv per year, and the highest single year of 25 mSv is below 50 mSv.

#22 · Medical Radiation Practitioner Core

A dose rate is 80 µGy per hour at 1 m from a small source. What is it at 4 m?

Answer

80 ÷ (4 × 4) = 80 ÷ 16 = 5 µGy per hour, using the inverse square law.

#23 · Medical Radiation Practitioner Core

A barrier is three half value layers thick and the unshielded rate would be 100 µSv per hour. What is the rate behind it?

Answer

Each half value layer halves the rate: 50, 25, then 12.5 µSv per hour.

#24 · Medical Radiation Practitioner Core

A technologist draws up a technetium-99m dose. Which protection is best, a lead apron or a syringe shield with forceps?

Answer

A syringe shield with forceps, plus short time and distance. A standard apron gives little protection against nuclear medicine gamma photons.

#25 · Medical Radiation Practitioner Core

Why is acrylic used before lead to shield strong beta emitters?

Answer

Dense metals such as lead make extra X-rays called bremsstrahlung when beta particles hit them, so a low atomic number material is used first.

#26 · Medical Radiation Practitioner Core

What is a diagnostic reference level (DRL)?

Answer

A dose value for a common examination on a typical sized patient, based on national data. It is a review tool for a department, not a dose limit for a patient.

#27 · Medical Radiation Practitioner Core

A department's typical dose for an examination is above the national DRL. What should it do?

Answer

Review its practice, such as protocol, collimation and extra views, correct what is excessive, re-measure and record. It should not stop the examination or blame one patient.

#28 · Medical Radiation Practitioner Core

A patient of childbearing potential may be pregnant before a pelvic examination. What is the best action?

Answer

Report to the referrer or responsible medical practitioner, who decides whether the examination is still justified, can wait, or can be replaced by ultrasound or MRI.

#29 · Medical Radiation Practitioner Core

A staff member declares she is pregnant. What is the correct approach to her work?

Answer

She tells her employer, the facility reviews her duties, and the foetal dose is kept to the lower value set by local rules. She is not automatically banned from all radiation work.

#30 · Medical Radiation Practitioner Core

What is a comforter or carer in radiation protection?

Answer

A person who is not staff and who knowingly and willingly helps in the care or comfort of a patient during an exposure. They are informed, protected and never forced.

#31 · Medical Radiation Practitioner Core

A mobile X-ray is under way and a relative asks to stay. What is the best first action?

Answer

Explain the radiation risk, offer protective clothing if the person is willing and helping is essential, keep them out of the primary beam and follow facility procedure.

#32 · Medical Radiation Practitioner Core

A radioactive liquid spills in the preparation room. What is the first thing to do?

Answer

Stop and warn others, then contain the spill with absorbent material from the edges inward while wearing gloves and a coat, and notify the radiation safety officer.

#33 · Medical Radiation Practitioner Core

What is the minimum number of patient identifiers used to confirm identity before a procedure in Australia?

Answer

At least three, such as full name, date of birth and one more item like the address or record number, checked against the request and wristband.

#34 · Medical Radiation Practitioner Core

A patient nods yes when you read out their name. Why is this a weak identity check?

Answer

Anxious, hard of hearing or confused patients may agree to anything. Ask them to state their own full name and date of birth instead.

#35 · Medical Radiation Practitioner Core

A woman of childbearing age is booked for a pelvic CT. What must you ask, and how?

Answer

Ask about the last menstrual period and possible pregnancy, privately and in plain words, and record the answer. Escalate if pregnancy is possible.

#36 · Medical Radiation Practitioner Core

The patient may be pregnant and the exam uses ionising radiation. What is the safest first action?

Answer

Pause and discuss with the referrer and supervising practitioner so justification is reviewed. An alternative without ionising radiation may be chosen.

#37 · Medical Radiation Practitioner Core

List the four features of valid consent.

Answer

It must be voluntary, informed, specific to the procedure and given by a person with capacity. It can be withdrawn at any time.

#38 · Medical Radiation Practitioner Core

What does capacity mean when deciding about consent?

Answer

The ability to understand the information, hold it long enough to weigh it up and communicate a choice. It is specific to the decision and the time.

#39 · Medical Radiation Practitioner Core

A relative says an unsure elderly patient will sign for him. What do you do?

Answer

Stop and assess whether he understands. If not, ask the referrer to arrange a capacity assessment and consent from the person with legal authority.

#40 · Medical Radiation Practitioner Core

Who should interpret during a consent discussion when the patient speaks little English?

Answer

A qualified interpreter, in person, by phone or by video. Never a child, and a relative is not a safe substitute. Document the use.

#41 · Medical Radiation Practitioner Core

Which vital sign is often the earliest and most reliable sign of deterioration, but most often not counted?

Answer

The respiratory rate. A rising rate can appear hours before other changes.

#42 · Medical Radiation Practitioner Core

What is the usual adult resting respiratory rate range and the SpO2 on room air?

Answer

About 12 to 20 breaths per minute, with SpO2 of 95 percent or higher on room air.

#43 · Medical Radiation Practitioner Core

An inpatient arriving for a scan is pale and sweaty with a breathing rate of 28 and pulse of 118. What is the best action?

Answer

Stop, stay with the patient, call for help using the local escalation or emergency number and hand over clearly. Do not finish the scan.

#44 · Medical Radiation Practitioner Core

What is the first-line drug for anaphylaxis and the route?

Answer

Adrenaline given intramuscularly into the outer mid-thigh. Antihistamines and steroids are not a substitute.

#45 · Medical Radiation Practitioner Core

What is the adrenaline dose for anaphylaxis in an adult and in a child?

Answer

Adults: 0.5 mg (0.5 mL of 1 mg per mL). Children: 0.01 mg per kg. It can be repeated after about five minutes if there is no improvement.

#46 · Medical Radiation Practitioner Core

A patient has an anaphylactic reaction on the table. Which position is correct?

Answer

Lie them flat, or sit up if breathing is hard, or on the left side if pregnant. Do not let them stand or walk.

#47 · Medical Radiation Practitioner Core

Give the DRSABCD steps and the CPR ratio.

Answer

Danger, Response, Send for help, Airway, Breathing, CPR, Defibrillation. CPR is 30 compressions to 2 breaths at 100 to 120 compressions per minute.

#48 · Medical Radiation Practitioner Core

Which precautions apply to every patient, and which are added for infections such as tuberculosis?

Answer

Standard precautions apply to every patient. For tuberculosis add airborne precautions: a P2 or N95 respirator and, where available, a negative pressure room.

#49 · Medical Radiation Practitioner Core

Which Domain 1 key capability of the Board's Professional capabilities covers medicines?

Answer

Key capability 9, safe and effective use of medicines. It covers knowing the medicine, the law that governs it and your scope of practice.

#50 · Medical Radiation Practitioner Core

Why are non-ionic iodinated contrast agents preferred for intravascular use?

Answer

They do not split into charged particles in solution and have lower osmolality, so they are better tolerated and less likely to cause side effects.

#51 · Medical Radiation Practitioner Core

A patient develops a few hives and mild itching after contrast. How is this graded and managed?

Answer

It is a mild reaction. Stay with the patient, reassure, observe, report to the radiologist and document, and advise about delayed reactions.

#52 · Medical Radiation Practitioner Core

How do you tell a vasovagal faint from anaphylaxis after an injection?

Answer

A faint gives a slow pulse and pale, sweaty skin. Anaphylaxis gives a fast weak pulse, hives, swelling and breathing difficulty and needs adrenaline.

#53 · Medical Radiation Practitioner Core

What does eGFR measure and why does it matter before contrast?

Answer

It estimates how well the kidneys filter blood, from blood creatinine. Contrast is cleared by the kidneys, so low eGFR raises the risk of acute kidney injury.

#54 · Medical Radiation Practitioner Core

The protocol needs a recent eGFR but there is none on record. What is the best first action?

Answer

Do not start the injection. Report the missing result to the radiologist, who decides whether to check kidney function or change the plan.

#55 · Medical Radiation Practitioner Core

A protocol gives 350 mgI/mL contrast, 90 mL. How many grams of iodine are injected?

Answer

350 mg/mL × 90 mL = 31,500 mg, which is 31.5 g of iodine.

#56 · Medical Radiation Practitioner Core

A protocol is 1.5 mL/kg with a maximum of 120 mL at 3 mL/s. How long is the injection for a 70 kg patient?

Answer

Volume is 1.5 × 70 = 105 mL, under the maximum. Time is 105 ÷ 3 = 35 seconds.

#57 · Medical Radiation Practitioner Core

During a power injection the patient reports pain and the forearm swells. What is the first action?

Answer

Stop the injection at once, then tell the radiologist, elevate the limb, assess circulation and document. Do not finish the run.

#58 · Medical Radiation Practitioner Core

What is nephrogenic systemic fibrosis (NSF)?

Answer

A rare, serious thickening of skin and organs seen in people with severe kidney impairment who received older, less stable gadolinium agents.

#59 · Medical Radiation Practitioner Core

When should barium sulfate be avoided?

Answer

When perforation is suspected, because leaked barium causes serious inflammation, and when there is a risk of aspiration into the lungs. Water-soluble contrast is used instead.

#60 · Medical Radiation Practitioner Core

Which six observations must a medical radiation practitioner know normal values for?

Answer

Respiratory rate, oxygen saturation, heart rate, blood pressure, temperature and level of consciousness.

#61 · Medical Radiation Practitioner Core

What is the first-line medicine for anaphylaxis and where is it given?

Answer

Adrenaline, into the muscle of the outer middle thigh, within scope and local protocol. Repeat after about 5 minutes if there is no improvement.

#62 · Medical Radiation Practitioner Core

What adrenaline dose per kilogram is used for a child with anaphylaxis, and what is it for 20 kg?

Answer

0.01 mg/kg, so 0.2 mg for 20 kg, which is 0.2 mL of a 1 mg/mL ampoule.

#63 · Medical Radiation Practitioner Core

Why should an anaphylaxis patient not be stood upright?

Answer

A sudden upright position can cause collapse. Lay the patient flat, or sit up only if breathing is the main problem.

#64 · Medical Radiation Practitioner Core

A sedated patient is lying in a quiet room. What must you keep doing?

Answer

Observe airway, breathing and level of consciousness, keep monitoring available and call for help early. Do not leave them unobserved.

#65 · Medical Radiation Practitioner Core

What is anatomical position, and why does it matter when describing left and right?

Answer

The person stands upright facing forward with arms by the sides and palms forward. Left and right always mean the patient's own left and right.

#66 · Medical Radiation Practitioner Core

Which plane divides the body into front and back?

Answer

The coronal plane. The sagittal plane divides left from right and the axial (transverse) plane divides upper from lower.

#67 · Medical Radiation Practitioner Core

On an axial CT slice, which side of the screen shows the patient's right?

Answer

The left side of the screen. Axial images are viewed as if standing at the patient's feet looking towards the head.

#68 · Medical Radiation Practitioner Core

What does the word part '-ectomy' mean, and what does 'brady-' mean?

Answer

'-ectomy' means surgical removal and 'brady-' means slow. Breaking a word into parts usually reveals its meaning.

#69 · Medical Radiation Practitioner Core

A patient on an erect chest image has a thin lung edge with no lung markings beyond it and is becoming very breathless with falling blood pressure. What is the best action?

Answer

Suspect a tension pneumothorax and alert the medical team at once. Do not wait for a formal report.

#70 · Medical Radiation Practitioner Core

What does a pleural effusion look like on an erect chest image?

Answer

Fluid blunts the costophrenic angle at the bottom of the lung and may show a curved fluid line. Fluid settles with gravity.

#71 · Medical Radiation Practitioner Core

A patient has sudden weakness of one arm and slurred speech. Which imaging is usually done first for a suspected stroke, and why?

Answer

A non-contrast CT of the brain, to rule out bleeding before treatment is chosen. Time matters because brain cells are lost every minute.

#72 · Medical Radiation Practitioner Core

What is the difference between a benign and a malignant tumour?

Answer

A benign tumour stays in one place and does not invade nearby tissue. A malignant tumour invades nearby tissue and can spread to distant sites.

#73 · Medical Radiation Practitioner Core

What do the letters T, N and M stand for in TNM staging?

Answer

T is the size and local extent of the primary tumour, N is lymph node involvement and M is distant metastasis.

#74 · Medical Radiation Practitioner Core

A report says a tumour is grade 3 and stage 1. What is the difference between grade and stage?

Answer

Grade describes how abnormal the cells look under the microscope. Stage describes how far the disease has spread.

#75 · Medical Radiation Practitioner Core

Name the three main ways cancer spreads to other parts of the body.

Answer

Through the lymphatic system to lymph nodes, through the blood to distant organs, and by direct local invasion.

#76 · Medical Radiation Practitioner Core

Why are fast-dividing tissues such as bone marrow and the gut lining affected earliest by radiation?

Answer

Cells that divide often and are less specialised are more radiosensitive, so damage shows up sooner in these tissues.

#77 · Medical Radiation Practitioner Core

A patient with a low eGFR is booked for a contrast CT. What should the practitioner do?

Answer

Follow the local protocol and refer the result to the radiologist or prescriber before giving contrast. Do not decide alone.

#78 · Medical Radiation Practitioner Core

A patient scheduled for an FDG PET has a high blood glucose. Why does this matter?

Answer

High blood glucose competes with the FDG tracer and reduces tumour uptake, so the result is checked before injection and the team is told.

#79 · Medical Radiation Practitioner Core

Why are children more at risk from radiation than adults, and what is the best practice?

Answer

Their cells divide rapidly and they have more years ahead for effects to appear. Use the lowest dose that gives an adequate image, with tight collimation and immobilisation.

#80 · Medical Radiation Practitioner Core

A patient in late pregnancy lying flat on her back says she feels faint. What is the safest action?

Answer

Tilt her onto her left side with support. The enlarged uterus can press on the inferior vena cava and reduce blood returning to the heart.

#81 · Professional and Ethical Practitioner

What does the Health Practitioner Regulation National Law do for medical radiation practice?

Answer

It sets up the national registration scheme, including the Medical Radiation Practice Board of Australia. Each state and territory has passed its own version, so rules are close to the same everywhere.

#82 · Professional and Ethical Practitioner

Which body decides registration standards and conduct outcomes: the Board or Ahpra?

Answer

The Board decides and sets standards. Ahpra is the agency that runs the register, receives notifications and carries out investigations for the Board.

#83 · Professional and Ethical Practitioner

Does registration with the Board replace a state licence to use ionising radiation?

Answer

No. State and territory radiation regulators license radiation sources and users under separate legislation. You usually need both.

#84 · Professional and Ethical Practitioner

What is a condition on registration?

Answer

A limit on how you practise, such as working only under supervision. You must follow every condition exactly, and practising outside one is a serious breach.

#85 · Professional and Ethical Practitioner

A manager asks a radiographer registered only in diagnostic radiography to cover a nuclear medicine shift. What should the radiographer do?

Answer

Decline and explain that their registration division does not cover nuclear medicine practice. Short staffing does not change the legal limit.

#86 · Professional and Ethical Practitioner

Which three standard questions test whether a task is within your scope?

Answer

Am I educated or trained to do it, am I competent to do it safely today, and is it allowed in this workplace and under the law? Any no means do not do it alone.

#87 · Professional and Ethical Practitioner

If you delegate a task to a trainee, who stays accountable for the decision to delegate?

Answer

You do. You must be sure the person is competent, the task is suitable and you give the right level of supervision.

#88 · Professional and Ethical Practitioner

Which patient needs the most direct supervision from a student: an anxious paediatric patient or a routine adult repeat?

Answer

The anxious paediatric patient. The higher the risk and the lower the experience, the more direct the supervision must be.

#89 · Professional and Ethical Practitioner

A trainee says a supervisor already checked a plan change but there is no record. What should the therapist do before signing it off?

Answer

Hold the change until the check is documented. Accountability stays with the person who signs, not with a verbal comment.

#90 · Professional and Ethical Practitioner

What is the legal status of the Board's Code of conduct?

Answer

It is not a law on its own, but it can be used in a disciplinary matter as evidence of proper professional behaviour.

#91 · Professional and Ethical Practitioner

Why does the practitioner carry the responsibility for keeping professional boundaries?

Answer

The therapeutic relationship is unequal, since the patient is often unwell, undressed and dependent. A patient saying they are comfortable does not remove the duty.

#92 · Professional and Ethical Practitioner

How do boundary problems usually begin?

Answer

In small steps, such as sharing too much personal information, giving out a private phone number, meeting outside work or favouring one patient.

#93 · Professional and Ethical Practitioner

Is a sexual or romantic relationship with a current patient ever acceptable?

Answer

No, never. A relationship with a former patient is also a serious concern if knowledge of their vulnerability could be used.

#94 · Professional and Ethical Practitioner

What is a conflict of interest?

Answer

A personal, professional or financial interest that could influence your judgement, such as treating a relative. Declare it and manage it openly.

#95 · Professional and Ethical Practitioner

A radiographer wants to post a scan on social media with no name attached. What is the problem?

Answer

Unique features in an image can still identify the patient. Posting anything that could identify a patient breaks confidentiality.

#96 · Professional and Ethical Practitioner

What are the four parts of valid informed consent?

Answer

The person has capacity, gives consent voluntarily, has enough information about purpose, risks and alternatives, and the consent covers the specific procedure.

#97 · Professional and Ethical Practitioner

Can a patient withdraw consent after a procedure has started?

Answer

Yes, at any time. You stop when consent is clearly withdrawn, unless stopping would create an immediate danger that you must manage safely.

#98 · Professional and Ethical Practitioner

What four abilities make up decision-making capacity?

Answer

Understanding the information, remembering it long enough to decide, weighing up the choices and communicating a decision.

#99 · Professional and Ethical Practitioner

Is a person assumed to have capacity?

Answer

Yes. Capacity is presumed unless there is good reason to doubt it, and a disability or an unwise choice does not by itself mean it is missing.

#100 · Professional and Ethical Practitioner

Is capacity the same for every decision a person makes?

Answer

No. It is specific to the decision and the time. A person may consent to a chest X-ray but not to a complex treatment plan.

#101 · Professional and Ethical Practitioner

An elderly patient with dementia refuses a needle but her daughter says to proceed. What is the best first action?

Answer

Assess capacity for this decision, check whether the daughter is a legally recognised substitute decision-maker, and tell the senior clinician. Do not override the patient on a relative's word.

#102 · Professional and Ethical Practitioner

Who can be a substitute decision-maker for a patient who lacks capacity?

Answer

It depends on state or territory law. Examples are a guardian, an enduring guardian, an attorney under an enduring power of attorney for health, or a person in a legal order of relationship.

#103 · Professional and Ethical Practitioner

What is an advance care directive?

Answer

A document or statement setting out a person's future health care wishes for when they lose capacity. A valid, applicable directive must be followed.

#104 · Professional and Ethical Practitioner

You are unsure whether an advance care directive applies to the current situation. What should you do?

Answer

Ask the senior clinician on duty, who can seek legal advice. Check it is current, applies to this situation and is presented by someone entitled to do so.

#105 · Professional and Ethical Practitioner

When may you share health information without the patient's consent?

Answer

When the law requires or permits it, for example a court order, a mandatory child protection report, or a serious and imminent threat to someone's life, health or safety.

#106 · Professional and Ethical Practitioner

A patient's employer phones to confirm an employee attended for an X-ray. What do you do?

Answer

Neither confirm nor deny. Even attendance is health information, so refer the caller to the patient.

#107 · Professional and Ethical Practitioner

A solicitor phones asking for a patient's results. What is the best response?

Answer

Do not give information over the phone. Follow the workplace process and refer the request to the person responsible for releasing records.

#108 · Professional and Ethical Practitioner

Is looking up a neighbour's record out of curiosity a breach if you tell nobody?

Answer

Yes. It is a serious privacy breach, and health services track who opens records.

#109 · Professional and Ethical Practitioner

What does the Notifiable Data Breaches scheme require?

Answer

Organisations must notify affected people and the Office of the Australian Information Commissioner when a breach is likely to cause serious harm. Report suspected breaches to your privacy officer.

#110 · Professional and Ethical Practitioner

What are the four grounds for a mandatory notification about a practitioner?

Answer

Practising while intoxicated, sexual misconduct, risk of substantial harm from an impairment, and risk of harm from a significant departure from accepted professional standards.

#111 · Professional and Ethical Practitioner

What threshold triggers a mandatory notification?

Answer

Forming a reasonable belief based on reasonable grounds. It is more than a rumour but you do not need proof.

#112 · Professional and Ethical Practitioner

Is having an impairment, by itself, a ground for a mandatory notification?

Answer

No. The duty is triggered when the impairment places the public at risk of substantial harm.

#113 · Professional and Ethical Practitioner

A technologist smells alcohol on a colleague about to start an anaesthetic list. What is the first priority, then the next step?

Answer

First keep the colleague out of patient care and tell the senior person in charge. Then make a mandatory notification, because a reasonable belief of practising while intoxicated has formed.

#114 · Professional and Ethical Practitioner

What is a voluntary notification?

Answer

A report about a concern that does not meet the mandatory grounds, such as poor communication or a breach of the Code. Anyone can make one.

#115 · Professional and Ethical Practitioner

Are people who notify honestly and in good faith protected?

Answer

Yes. The National Law protects them from civil, criminal and administrative liability.

#116 · Professional and Ethical Practitioner

What are the four things a patient must show to succeed in negligence?

Answer

A duty of care was owed, the standard of care was not met, the breach caused the harm, and the harm was a real loss.

#117 · Professional and Ethical Practitioner

What standard of care is applied in negligence?

Answer

What a reasonable practitioner of the same kind, with the same training, would have done.

#118 · Professional and Ethical Practitioner

What is open disclosure?

Answer

A prompt, honest and compassionate conversation with the patient and family about what happened, with regret expressed, what is known, what will prevent a repeat and what support is offered.

#119 · Professional and Ethical Practitioner

A radiographer realises the wrong limb was imaged. What is the best action?

Answer

Tell the supervisor straight away, report the incident, take part in open disclosure and arrange any repeat appropriately. Do not quietly repeat the correct side.

#120 · Professional and Ethical Practitioner

Which Australian framework guides open disclosure?

Answer

The Australian Open Disclosure Framework from the Australian Commission on Safety and Quality in Health Care.

#121 · Professional and Ethical Practitioner

Name the four principles of biomedical ethics in the Professional capabilities.

Answer

Autonomy, beneficence, non-maleficence and justice, from Beauchamp and Childress.

#122 · Professional and Ethical Practitioner

A capable adult refuses a treatment you believe would help. Which principles clash and what do you do?

Answer

Autonomy clashes with beneficence. Make sure the refusal is informed, document it and keep the door open to further care.

#123 · Professional and Ethical Practitioner

What does cultural safety mean for Aboriginal and Torres Strait Islander patients?

Answer

Care that they experience as safe, defined by them and not by the practitioner. It goes beyond cultural awareness.

#124 · Professional and Ethical Practitioner

For an important consent discussion with a patient who speaks little English, who should interpret?

Answer

An accredited interpreter, not a family member.

#125 · Professional and Ethical Practitioner

You have a reasonable concern about a child's safety. What is the best action?

Answer

Follow your workplace child protection procedure and report to the right authority or to the senior person who will do so. Mandatory reporting rules differ by state and territory.

#126 · Communicator and Collaborator

What is health literacy?

Answer

A person's ability to find, understand and use health information. It is not the same as intelligence, and stress can lower it.

#127 · Communicator and Collaborator

What is teach-back and why is it used?

Answer

You ask the patient to explain the information in their own words. It checks understanding, and a gap means your explanation needs work.

#128 · Communicator and Collaborator

A patient nods during your explanation of a procedure. What is the best way to check they understood?

Answer

Use teach-back: ask them to describe in their own words what will happen. A nod or a yes to 'Do you understand?' is not reliable.

#129 · Communicator and Collaborator

What is consent, according to the Board's explanatory notes?

Answer

A person's voluntary decision about healthcare made with knowledge and understanding of the benefits and risks involved.

#130 · Communicator and Collaborator

During treatment a patient says 'stop'. What is the safest action?

Answer

Pause and talk with the patient. Consent can be withdrawn at any time, so you do not finish the last few seconds.

#131 · Communicator and Collaborator

A patient has signed the consent form but looks confused about the procedure. What should you do?

Answer

Explain again in plain language and check with teach-back. A signature without understanding is not valid consent.

#132 · Communicator and Collaborator

Why should you keep communicating with a patient during an examination or treatment?

Answer

To confirm ongoing consent, reassure the patient and notice distress or non-verbal cues early.

#133 · Communicator and Collaborator

What is a CALD patient?

Answer

A person from a culturally and linguistically diverse background, whose cultural identity or language differs from the main culture of the setting.

#134 · Communicator and Collaborator

When should you use a qualified interpreter?

Answer

Whenever a patient's first language is not English and communication may be limited, including for consent and sensitive questions.

#135 · Communicator and Collaborator

A family member offers to interpret for an elderly patient. What is the best action?

Answer

Politely arrange a qualified interpreter, because relatives may leave out or soften information and privacy and accuracy are at risk.

#136 · Communicator and Collaborator

Can a child be used to interpret for a parent?

Answer

No. A child must never interpret for a parent or relative in a health setting.

#137 · Communicator and Collaborator

In what situation might you use the best available means of communication instead of a qualified interpreter?

Answer

A genuine emergency where no interpreter is available and delay would cause harm. You document what was done.

#138 · Communicator and Collaborator

What should be recorded in the notes when an interpreter is used?

Answer

That an interpreter was used, their name or ID and how they attended (in person, phone or video).

#139 · Communicator and Collaborator

A patient asks you 'Is it cancer?' after a scan. What is the best response?

Answer

Be honest and kind: acknowledge the worry, do not guess a diagnosis, and explain who will give the results and when.

#140 · Communicator and Collaborator

Why is 'I'm sure it's nothing' a poor reply to a worried patient awaiting results?

Answer

It is false reassurance and may be wrong. Staying inside your scope while being warm and clear about who will explain is better.

#141 · Communicator and Collaborator

What does cultural safety mean?

Answer

A person feels safe and respected in receiving care, free of racism and discrimination, and the patient decides whether care was culturally safe.

#142 · Communicator and Collaborator

How does cultural safety differ from cultural competence?

Answer

Cultural competence is knowing and applying cultural knowledge. Cultural safety adds reflection on the practitioner's own bias and power, judged by the patient.

#143 · Communicator and Collaborator

Put these in order from least to most advanced: cultural safety, cultural awareness, cultural competence, cultural sensitivity.

Answer

Cultural awareness, cultural sensitivity, cultural competence, then cultural safety.

#144 · Communicator and Collaborator

According to the Board's Domain 3, what must practitioners reflect on about themselves?

Answer

Their own culture and how it shapes their perceptions of and interactions with others, including the impact of systemic racism and bias.

#145 · Communicator and Collaborator

What historical events help explain mistrust of health services among some Aboriginal and Torres Strait Islander Peoples?

Answer

Dispossession of land, forced removal of children (the Stolen Generations) and past health care that was harmful or disrespectful.

#146 · Communicator and Collaborator

Why can treatment far from home be hard for an Aboriginal or Torres Strait Islander patient?

Answer

Connection to Country, family and community is central to wellbeing, so being away for weeks of treatment can be a heavy burden.

#147 · Communicator and Collaborator

An Aboriginal patient misses a second treatment appointment. What is the best action?

Answer

Ask what the barrier is (transport, accommodation, family obligations, fear), involve the Aboriginal Liaison Officer and plan together, rather than recording non-compliance.

#148 · Communicator and Collaborator

What can silence from a patient mean in some cultural settings?

Answer

It may mean the person is thinking, not agreeing or refusing. Allow time rather than filling the gap.

#149 · Communicator and Collaborator

Who can support cultural safety and communication for Aboriginal and Torres Strait Islander patients in hospital?

Answer

Aboriginal Liaison Officers, Aboriginal Health Workers and Aboriginal Health Practitioners, involved early with the patient's agreement.

#150 · Communicator and Collaborator

A female patient prefers a female practitioner for a sensitive examination. What is the best response?

Answer

Meet the request where reasonably possible or offer the earliest alternative, while protecting privacy and dignity with a gown or drape.

#151 · Communicator and Collaborator

A transgender patient tells you their name and pronouns. What do you do?

Answer

Use them, and correct yourself if you slip. Ask only for information needed for safe care, privately and with a clear reason.

#152 · Communicator and Collaborator

A patient with a hearing impairment arrives with a support person. Who do you speak to?

Answer

The patient, with adjustments such as a hearing loop, Auslan interpreter or written information, not only the support person.

#153 · Communicator and Collaborator

What are the principles of trauma-informed care?

Answer

Safety, trust, choice, collaboration and empowerment.

#154 · Communicator and Collaborator

A patient freezes and becomes distressed during positioning. What is the best action?

Answer

Pause and ask what would help. Do not push on, and do not ask for details of past trauma unless needed for care.

#155 · Communicator and Collaborator

What does ISBAR stand for?

Answer

Identify, Situation, Background, Assessment, Request or Recommendation.

#156 · Communicator and Collaborator

In ISBAR, what belongs under Background?

Answer

Relevant history, allergies, medicines, cannulas, infection status, mobility and language needs.

#157 · Communicator and Collaborator

What is the minimum number of approved patient identifiers expected at handover?

Answer

At least three approved patient identifiers.

#158 · Communicator and Collaborator

Which NSQHS Standard covers clinical handover and correct patient identification?

Answer

The Communicating for Safety standard (Standard 6).

#159 · Communicator and Collaborator

A patient has a reaction after contrast and you call the ward. What is the best way to structure the call?

Answer

Use ISBAR: identify yourself and the patient, state the situation, give allergy and history, give your assessment and state exactly what you need.

#160 · Communicator and Collaborator

You see a risk that a senior colleague seems to ignore. What is the best action?

Answer

Speak up clearly, state your concern and what you need, and escalate up the chain if it is not addressed. Staying silent is the wrong choice.

#161 · Communicator and Collaborator

What does CUS stand for in escalation language?

Answer

I am Concerned, I am Uncomfortable, this is a Safety issue.

#162 · Communicator and Collaborator

You delegate a task to a student. Who remains accountable for the patient?

Answer

The supervising practitioner. You delegate only what the person is trained and authorised to do and you check the result.

#163 · Communicator and Collaborator

A student has not yet been signed off to perform a task. What should you do?

Answer

Supervise them directly or do the task yourself. Do not leave them alone with the patient for it.

#164 · Communicator and Collaborator

You made an error in a clinical note. What is the correct way to fix it?

Answer

Correct it by the service's process so the original stays visible, and add a dated correction. Never delete or alter the original.

#165 · Communicator and Collaborator

What makes good documentation?

Answer

It is accurate, factual, timely, clear, identifiable, complete and confidential, and it records consent and interpreter use.

#166 · Lifelong Learner

What does evidence-based practice join together?

Answer

The best available research evidence, the practitioner's clinical expertise and context, and the patient's values and preferences.

#167 · Lifelong Learner

Which study designs sit highest in the evidence hierarchy for treatment questions?

Answer

Systematic reviews and meta-analyses, then randomised controlled trials. Cohort, case-control, case reports and expert opinion sit lower.

#168 · Lifelong Learner

In a test, 90 of 100 people with disease test positive. What is the sensitivity?

Answer

Sensitivity = true positives ÷ all with disease = 90 ÷ 100 = 90%. The test rarely misses disease.

#169 · Lifelong Learner

Standard care gives a severe skin reaction in 8 of 100 patients and a new routine in 4 of 100. What is the relative risk?

Answer

Relative risk = 0.04 ÷ 0.08 = 0.5. The new routine halves the risk, an absolute difference of 4 in every 100.

#170 · Lifelong Learner

A study reports a relative risk with a confidence interval that includes 1. What can you conclude?

Answer

The study has not shown a clear difference between the groups. A relative risk of 1 means no difference.

#171 · Lifelong Learner

A new study has a small p-value. Does that prove the treatment gives a large benefit?

Answer

No. A small p-value only suggests chance alone is unlikely. It does not show the effect is large or important.

#172 · Lifelong Learner

Which kind of test is best for screening, and which for confirming a diagnosis before major treatment?

Answer

Screening needs high sensitivity so few cases are missed. Confirming a diagnosis needs high specificity so healthy people are not treated by mistake.

#173 · Lifelong Learner

A department checks whether patient identification was recorded for every examination in a month. Is this research?

Answer

No. Measuring your own service against its existing standard is audit or quality improvement. Testing something new on patients or producing generalisable findings is research needing HREC review.

#174 · Lifelong Learner

What are the four steps of the quality improvement cycle, in order?

Answer

Plan, do, study, act. Identify the problem, try a small change, measure the result, then adopt, adjust or drop the change and repeat.

#175 · Lifelong Learner

You are rostered to a study you have only watched once. What is the safest action?

Answer

Tell the senior technologist or supervisor you are not signed off, ask for supervision or reassignment, and arrange training. Do not proceed quietly or refuse without telling anyone.

#176 · Lifelong Learner

Is CPD a registration duty, and what must it include?

Answer

Yes. It must be relevant to your scope, planned, recorded as evidence for audit, and include ongoing cultural learning. Check the Board's current standard for the hours.

#177 · Lifelong Learner

Describe good feedback to a student after a procedure.

Answer

Timely, specific, balanced, focused on behaviour and not the person, and followed by a plan for the next step.

#178 · Lifelong Learner

A student is about to perform a procedure beyond their level. Who stays responsible, and what should happen?

Answer

The supervising practitioner stays responsible for patient safety. Supervision must match the learner's level, and the patient must be told a student is involved.

#179 · Lifelong Learner

An AI tool auto-draws a radiotherapy contour that does not match the anatomy. What do you do?

Answer

Do not use it. Check it against the anatomy, correct or redo it, and report the problem through the incident system. AI is decision support, not a decision.

#180 · Lifelong Learner

What does random allocation do to make a trial more trustworthy?

Answer

It spreads known and unknown differences (confounders) evenly between groups, so a difference in outcome is more likely due to the treatment.

#181 · Safety Quality and Risk Management

How many key capabilities does Domain 5 of the Board's professional capabilities contain, and what is its theme?

Answer

Four: safe radiation practice, patient safety, quality assurance, and a safe workplace. Its theme is protecting patients and others from harm.

#182 · Safety Quality and Risk Management

What is a near miss in healthcare safety?

Answer

An event that could have caused harm but was caught before it reached the patient, such as spotting a wrong name on a request before the exposure.

#183 · Safety Quality and Risk Management

What is a sentinel event?

Answer

A rare, serious event causing death or severe harm that is on the national list kept by the ACSQHC. Some services call a preventable one a never event.

#184 · Safety Quality and Risk Management

A wrong-patient mix-up happens at handover. Which NSQHS Standard is it mainly about?

Answer

Communicating for Safety. Handover and patient identification failures sit under this standard.

#185 · Safety Quality and Risk Management

Name the eight NSQHS Standards.

Answer

Clinical Governance, Partnering with Consumers, Preventing and Controlling Healthcare-Associated Infection, Medication Safety, Comprehensive Care, Communicating for Safety, Blood Management, and Recognising and Responding to Acute Deterioration.

#186 · Safety Quality and Risk Management

A patient has an unrecognised contrast reaction. Which NSQHS Standard is most relevant?

Answer

Recognising and Responding to Acute Deterioration.

#187 · Safety Quality and Risk Management

What does root cause analysis ask?

Answer

Why a serious event happened, not who to blame. It usually finds system causes and leads to system fixes.

#188 · Safety Quality and Risk Management

What are the elements of open disclosure?

Answer

Express regret, give the facts as known, explain what is being done, and describe what will be done to prevent it happening again.

#189 · Safety Quality and Risk Management

You notice a mistake happened on a patient. What is your first step after making them safe?

Answer

Report it through the incident system and tell your supervisor so open disclosure and review are done properly.

#190 · Safety Quality and Risk Management

What is the idea behind a Just Culture in safety reporting?

Answer

Honest mistakes and system faults lead to learning and support, while reckless shortcuts are treated differently, so staff feel safe to report.

#191 · Safety Quality and Risk Management

You skipped checking the wristband but the patient's name and date of birth matched. No harm occurred. What should you do?

Answer

Report it as an incident or near miss and tell your supervisor, because the process failed even though no harm resulted.

#192 · Safety Quality and Risk Management

When must a practitioner make a mandatory notification about another registered health practitioner?

Answer

When they form a reasonable belief the person has placed the public at risk of substantial harm, for example through impairment, intoxicated practice, a significant departure from professional standards or sexual misconduct.

#193 · Safety Quality and Risk Management

Telling your manager about a colleague's unsafe practice has been done. Does the mandatory notification duty end?

Answer

No. The legal duty to notify under the National Law remains if the grounds are met, even though a manager has been told.

#194 · Safety Quality and Risk Management

How is a risk score calculated in a risk matrix?

Answer

Likelihood × consequence.

#195 · Safety Quality and Risk Management

A hazard has likelihood 3 and consequence 4. What is the risk score?

Answer

3 × 4 = 12.

#196 · Safety Quality and Risk Management

List the hierarchy of controls from most to least effective.

Answer

Eliminate, substitute, engineering controls, administrative controls, then personal protective equipment.

#197 · Safety Quality and Risk Management

A stockroom floor is often wet and slippery. Which control is higher in the hierarchy: fixing the leak or putting up a warning sign?

Answer

Fixing the leak. That is eliminating the hazard, while a sign is only an administrative control.

#198 · Safety Quality and Risk Management

Which controls does the Board state must be included for radiation protection?

Answer

Time, distance and patient shielding.

#199 · Safety Quality and Risk Management

A source gives 8 µSv/h at 1 m. What is the dose rate at 2 m?

Answer

$8 \div 2^2 = 8 \div 4 = 2$ µSv/h, by the inverse square law.

#200 · Safety Quality and Risk Management

A source gives 8 µSv/h at 1 m. What is the dose rate at 4 m?

Answer

$8 \div 4^2 = 8 \div 16 = 0.5$ µSv/h.

#201 · Safety Quality and Risk Management

At 6 µSv/h, what dose is received in 15 minutes?

Answer

15 minutes is 0.25 h, so 6 × 0.25 = 1.5 µSv.

#202 · Safety Quality and Risk Management

What do justification, limitation and optimisation mean?

Answer

Justification: the exposure does more good than harm. Limitation: risk is kept within acceptable limits. Optimisation: exposure is as low as reasonably achievable (ALARA).

#203 · Safety Quality and Risk Management

What is the occupational dose limit used in Australia?

Answer

20 mSv per year averaged over five years, with no more than 50 mSv in any single year.

#204 · Safety Quality and Risk Management

What is the annual dose limit for members of the public?

Answer

1 mSv per year.

#205 · Safety Quality and Risk Management

Do dose limits apply to a patient having a justified medical exposure?

Answer

No. Patients are protected through justification, optimisation and diagnostic reference levels instead.

#206 · Safety Quality and Risk Management

Which agency publishes the Radiation Protection Series, including the code for the medical use of radiation?

Answer

The Australian Radiation Protection and Nuclear Safety Agency (ARPANSA).

#207 · Safety Quality and Risk Management

Why might the radiation regulator and forms differ if you move interstate?

Answer

Radiation safety law is made by states and territories, so regulators and some requirements differ between jurisdictions.

#208 · Safety Quality and Risk Management

A request seems to be for the wrong side or an unnecessary repeat. What should you do?

Answer

Pause and contact the referrer to clarify before performing it.

#209 · Safety Quality and Risk Management

Who should advise before you change radiation dose settings away from protocol?

Answer

A qualified expert such as the radiation safety officer. Do not change settings on your own.

#210 · Safety Quality and Risk Management

What is the minimum number of patient identifiers the Board expects?

Answer

At least three recognised identifiers, such as full name, date of birth and record number.

#211 · Safety Quality and Risk Management

Why should you ask the patient to state their name rather than read it out and ask 'is that you'?

Answer

A confused or anxious patient may agree to the wrong name. Ask them to state name and date of birth, then compare with request and wristband.

#212 · Safety Quality and Risk Management

How do you identify an unconscious patient?

Answer

Use the wristband and the transferring staff member or carer, and follow the local identification procedure.

#213 · Safety Quality and Risk Management

What is a time-out?

Answer

A deliberate pause just before a procedure or treatment where the team confirms patient, procedure, site and side together.

#214 · Safety Quality and Risk Management

A junior team member says 'stop' during a time-out because the side marker looks wrong. How should this be treated?

Answer

As a good catch. Anyone on the team can call a stop, and the discrepancy is resolved before continuing.

#215 · Safety Quality and Risk Management

What are the five rights used for medication safety checks?

Answer

Right patient, drug, dose, route and time, plus allergy status and expiry and labelling.

#216 · Safety Quality and Risk Management

You find an unlabelled syringe on the tray. What do you do?

Answer

Do not use it. Discard or return it per local policy and prepare a correctly labelled one.

#217 · Safety Quality and Risk Management

A patient starts having a reaction during an injection. What is the first action?

Answer

Stop the injection, call for help, follow the local acute reaction plan and document.

#218 · Safety Quality and Risk Management

What does the infection chain need?

Answer

A host, an agent, and a way for the agent to pass from one to the other.

#219 · Safety Quality and Risk Management

What are the 5 Moments for Hand Hygiene?

Answer

Before touching a patient, before a procedure, after a procedure or body fluid exposure risk, after touching a patient, and after touching patient surroundings.

#220 · Safety Quality and Risk Management

When should you use soap and water instead of alcohol-based hand rub?

Answer

When hands are visibly soiled, or after contact with certain infections such as Clostridioides difficile.

#221 · Safety Quality and Risk Management

Do gloves replace hand hygiene?

Answer

No. Gloves are worn for blood or body fluid contact, and hand hygiene is still needed, with gloves changed between patients.

#222 · Safety Quality and Risk Management

What do you do immediately after a needlestick injury?

Answer

Wash the site, report at once, and follow the exposure management plan.

#223 · Safety Quality and Risk Management

Should used needles be recapped by hand?

Answer

No. Place them in a sharps container at the point of use, and do not overfill it.

#224 · Safety Quality and Risk Management

Which guidelines are the basis for infection control in Australian healthcare?

Answer

The NHMRC Australian Guidelines for the Prevention and Control of Infection in Healthcare (2019).

#225 · Safety Quality and Risk Management

What primary duty does WHS law place on the person conducting a business or undertaking?

Answer

To keep workers and others safe so far as reasonably practicable.

#226 · Safety Quality and Risk Management

What are a worker's WHS duties?

Answer

Take reasonable care for their own safety and others, follow reasonable instructions, and cooperate with safety policies.

#227 · Safety Quality and Risk Management

Is the MRI magnet only a hazard while scanning?

Answer

No. The magnet is always on, so projectile risk exists at all times.

#228 · Safety Quality and Risk Management

A patient collapses in the MRI scanner room. What is the first priority?

Answer

Move the patient out of the magnet room to a safe area, and the resuscitation team does not enter the magnet room.

#229 · Safety Quality and Risk Management

A test on a treatment machine fails. What should you do?

Answer

Take it out of use, label and report it, and do not treat patients until it is fixed and passes the checks.

#230 · Safety Quality and Risk Management

What do the letters in RACE stand for in a fire?

Answer

Remove people in danger, raise the Alarm, Contain the fire by closing doors, then Extinguish if safe or Evacuate.

#231 · Leader and Steward

According to the Board's Leader and Steward domain, who is expected to show leadership?

Answer

Every medical radiation practitioner with general registration, not only people in formal management roles.

#232 · Leader and Steward

Name the three key capabilities in the Board's Leader and Steward domain.

Answer

Contribute to improving healthcare delivery in teams, organisations and systems; lead and support stewardship of healthcare systems; and leadership as a professional practice.

#233 · Leader and Steward

What does stewardship mean in health care?

Answer

Looking after shared, limited things such as money, staff time, equipment, the environment and public trust so that they serve patients well now and later.

#234 · Leader and Steward

What does a practitioner remain responsible for after delegating a task?

Answer

Accountability for the decision that the delegation was safe, which means the task matched the person's competence, scope and level of supervision.

#235 · Leader and Steward

A busy department asks a first-year student to position a patient alone for a repeat examination. What should you do?

Answer

Decline politely, explain that the student needs supervision for this task, and supervise it yourself or find another way to share the load. Busyness does not lower the supervision standard.

#236 · Leader and Steward

What three questions guide a safe delegation?

Answer

Is the person competent for the task, is it within their scope of practice or role, and is the supervision matched to the risk?

#237 · Leader and Steward

Three requests arrive at once: a routine ward patient, an emergency patient with a suspected serious injury, and an outpatient waiting 40 minutes. Who is seen first?

Answer

The emergency patient, because prioritisation follows patient risk and urgency. Tell the outpatient honestly about the delay and inform the coordinator.

#238 · Leader and Steward

What should never be shortened when you are under time pressure?

Answer

Core safety steps such as patient identification, the procedure check, radiation safety and infection control.

#239 · Leader and Steward

A shift is going to be unsafe because too few trained staff are rostered. What is the leadership action?

Answer

Tell the coordinator early, state what is at risk and suggest a solution. Reporting unsafe staffing is a leadership act.

#240 · Leader and Steward

What is low-value care?

Answer

Tests or treatments that give little or no benefit to the patient, or where the harm or cost outweighs the benefit.

#241 · Leader and Steward

A referral asks for a repeat examination two days after a clear one with no new symptoms. What is the best first action?

Answer

Contact the referrer, ask what has changed and share the findings of the recent study, then agree on the next step and document the discussion.

#242 · Leader and Steward

What is Choosing Wisely Australia?

Answer

A national initiative that encourages clinicians and patients to talk about care that may not be needed. It supports questioning a request but does not replace clinical judgement.

#243 · Leader and Steward

Which level of government is mainly responsible for Medicare, the MBS and the PBS?

Answer

The Commonwealth (national) government.

#244 · Leader and Steward

Who mainly runs public hospitals and holds the main radiation safety legislation powers in Australia?

Answer

State and territory governments, which is why radiation rules can differ in detail between jurisdictions.

#245 · Leader and Steward

What does the Medicare Benefits Schedule list?

Answer

The services that attract a Medicare rebate, including many diagnostic imaging services.

#246 · Leader and Steward

A patient asks you for the exact rebate amount for their scan and you are unsure. What is the best response?

Answer

Explain that funding differs between services and refer the question to the administration team, rather than guessing a figure.

#247 · Leader and Steward

What is a near miss and why should it be reported?

Answer

An event that could have caused harm but did not. Reporting it reveals weaknesses before a patient is hurt.

#248 · Leader and Steward

Which framework guides honest communication with a patient and family after harm occurs?

Answer

The Australian Open Disclosure Framework: acknowledge, express regret, explain what is known, describe what will be done and keep them informed.

#249 · Leader and Steward

What are the four steps of the plan, do, study, act cycle?

Answer

Plan the change and how to measure it, do it on a small scale, study the result against the plan, then act by keeping, changing or dropping it.

#250 · Leader and Steward

A senior colleague starts a procedure without the final safety check. What is the best action?

Answer

Pause the team and ask in a calm, clear voice for the check to be completed. Do not assume the more senior person must be right.

#251 · Diagnostic Radiography Division

What does an anteroposterior (AP) projection mean for beam direction through the patient?

Answer

The beam enters the front of the body and leaves the back. A PA projection is the reverse.

#252 · Diagnostic Radiography Division

Why is a part placed as close to the image receptor as possible?

Answer

A short object to image distance gives less magnification and better sharpness. A longer distance magnifies and blurs the structure.

#253 · Diagnostic Radiography Division

What is the rule for imaging a bone or joint to avoid missing a fracture or foreign body?

Answer

Take at least two projections at right angles to each other. A single view can hide a fracture lying in the line of the beam.

#254 · Diagnostic Radiography Division

When must the anatomical side marker be placed?

Answer

Within the primary beam and the collimated field, on the correct side, before the exposure. Never guess a side or rely on adding it afterwards.

#255 · Diagnostic Radiography Division

A side marker was forgotten and the side cannot be established with certainty. What should you do?

Answer

Repeat the exposure with the correct marker. A wrong side on an image is a patient safety risk.

#256 · Diagnostic Radiography Division

What is the routine chest projection and the usual SID?

Answer

An erect PA at full inspiration, usually at 180 cm SID, which reduces magnification of the heart.

#257 · Diagnostic Radiography Division

Which features show a well taken PA chest image?

Answer

About 10 posterior ribs above the diaphragm, no rotation (clavicle ends equal from the spinous processes), vertebrae just visible through the heart, and scapulae clear of the lungs.

#258 · Diagnostic Radiography Division

Why does the heart look larger on an AP supine chest than on an erect PA?

Answer

The heart is further from the receptor, so it is magnified. Supine fluid also layers behind the lung as a general haze.

#259 · Diagnostic Radiography Division

A patient cannot sit up and a small pneumothorax is suspected. Which view can help?

Answer

A lateral decubitus with a horizontal beam, because free air rises to the highest point.

#260 · Diagnostic Radiography Division

Which views show free gas in a patient with a suspected perforated bowel?

Answer

An erect chest film or a left lateral decubitus abdomen (right side up) with a horizontal beam. The patient lies in position first.

#261 · Diagnostic Radiography Division

What does free gas look like on an erect film?

Answer

A thin lucent crescent beneath the diaphragm.

#262 · Diagnostic Radiography Division

A trauma patient has a suspected neck injury. What is the first cervical spine view and how is it taken?

Answer

A cross table lateral with a horizontal beam, taken with the patient supine and not moved.

#263 · Diagnostic Radiography Division

What does cross table mean in radiography?

Answer

The beam is horizontal, parallel to the floor, so the patient can stay flat on the trolley.

#264 · Diagnostic Radiography Division

Which extra cervical view shows the atlas and axis?

Answer

The open mouth view, which shows C1 and C2.

#265 · Diagnostic Radiography Division

A patient with a suspected fractured hip has a shortened, externally rotated leg. How should the AP pelvis be taken?

Answer

Take it as the patient lies, without rotating the leg, because rotation could move the fragments.

#266 · Diagnostic Radiography Division

Which pair of views is usual for a suspected fractured neck of femur?

Answer

An AP pelvis and a cross table lateral of the affected hip, with the other leg lifted out of the way.

#267 · Diagnostic Radiography Division

In a routine AP pelvis, how are the legs normally positioned and why?

Answer

Internally rotated about 15° so the femoral necks are seen in profile, unless a fracture is suspected.

#268 · Diagnostic Radiography Division

Why is a scaphoid series taken with ulnar deviation, and why is it worth repeating films?

Answer

Ulnar deviation elongates the scaphoid and separates it from the surrounding carpals. A scaphoid fracture can be invisible on the first films, so if the films are negative but the clinical suspicion remains, the wrist is immobilised and repeat films or other imaging are done after about 7 to 14 days.

#269 · Diagnostic Radiography Division

What does a raised fat pad sign at the elbow suggest?

Answer

A joint effusion, often from an occult fracture, especially in children.

#270 · Diagnostic Radiography Division

Why is a second view added to an AP shoulder?

Answer

A Y view or axial view stops a dislocation being missed. An AP alone can miss a posterior dislocation.

#271 · Diagnostic Radiography Division

What are the three routine ankle views, and how is the mortise view positioned?

Answer

AP, mortise and lateral. For the mortise the leg is internally rotated about 15° to 20° to open the joint space.

#272 · Diagnostic Radiography Division

Which knee view detects a lipohaemarthrosis, and what does it mean?

Answer

A horizontal beam lateral. A fat and blood level in the joint means an intra articular fracture until proven otherwise.

#273 · Diagnostic Radiography Division

In a panoramic OPG, what happens if the patient's chin is too high?

Answer

The occlusal plane curves the wrong way and the incisors look blurred. Positioning errors are the usual cause of OPG faults.

#274 · Diagnostic Radiography Division

Why should earrings and necklaces be removed before an OPG?

Answer

Metal creates dense artefacts that hide anatomy.

#275 · Diagnostic Radiography Division

A parent helps hold a child during an exposure. What must they do?

Answer

Wear a lead apron, stay out of the primary beam and not be pregnant. Immobilisation devices are used first, hand holding last.

#276 · Diagnostic Radiography Division

A radiographer suspects non accidental injury in a child. What is the correct action?

Answer

Follow the skeletal survey protocol and local policy, and report concerns through the correct pathway. Do not question the child or carer.

#277 · Diagnostic Radiography Division

Which three questions decide whether an image should be repeated?

Answer

Is the anatomy of interest shown, is positioning good enough to answer the clinical question, and is the exposure adequate? Repeat only when the image cannot answer the question.

#278 · Diagnostic Radiography Division

An image looks fine but the exposure index is very high. What does that mean?

Answer

The image was overexposed and the patient received extra dose. Record it and adjust technique next time. A repeat is rarely justified.

#279 · Diagnostic Radiography Division

What does kVp control in the x-ray beam, and how does raising it change the image?

Answer

kVp sets photon energy, so it controls penetration and contrast. Higher kVp gives a more penetrating beam, lower contrast and lower patient dose for the same detector signal.

#280 · Diagnostic Radiography Division

What does mAs control, and what happens to dose if you double it?

Answer

mAs sets the number of photons made. Doubling mAs doubles the amount of radiation and so doubles the patient dose.

#281 · Diagnostic Radiography Division

State the 15 percent rule.

Answer

Raising kVp by 15 percent roughly doubles the exposure reaching the detector, and lowering it by 15 percent roughly halves it. To keep the same signal after a rise, halve the mAs.

#282 · Diagnostic Radiography Division

A hand is exposed at 70 kVp and 20 mAs. Using the 15 percent rule, what factors give the same detector signal at lower dose?

Answer

70 × 1.15 = 80.5, so use about 80 kVp, and halve the mAs from 20 to 10. Signal stays similar, dose falls and contrast is slightly lower.

#283 · Diagnostic Radiography Division

An exposure of 10 mAs at 100 cm is changed to 150 cm SID. What new mAs keeps the same detector signal?

Answer

New mAs = 10 × (150 ÷ 100)² = 10 × 2.25 = 22.5 mAs. A longer distance weakens the beam, so mAs must rise.

#284 · Diagnostic Radiography Division

What does the inverse square law say about beam intensity and distance?

Answer

Intensity is inversely proportional to the square of the distance from the source. Doubling the distance cuts intensity to one quarter.

#285 · Diagnostic Radiography Division

Which end of the tube should lie over the thicker part of the body, and why?

Answer

The cathode end, because the anode heel effect makes the beam less intense on the anode side. For an AP thoracic spine the cathode goes at the feet end.

#286 · Diagnostic Radiography Division

A thoracic spine image is too dark at the top and too light at the bottom. What is the best first correction?

Answer

Reverse the tube direction so the cathode is over the thicker lower spine, using the heel effect. Raising mAs would darken everything and double the dose.

#287 · Diagnostic Radiography Division

How do you work out magnification from the geometry?

Answer

Magnification = SID ÷ SOD. With SID 100 cm and OID 10 cm, SOD is 90 cm, so magnification is 1.11 and a 5.0 cm object appears about 5.6 cm.

#288 · Diagnostic Radiography Division

State the formula for geometric unsharpness (penumbra).

Answer

Penumbra = focal spot size × OID ÷ SOD. A focal spot of 1.0 mm, OID 10 cm and SOD 90 cm gives 0.11 mm.

#289 · Diagnostic Radiography Division

Which three changes reduce geometric unsharpness?

Answer

Use a smaller focal spot, a longer SID, and place the part as close to the detector as possible (smaller OID).

#290 · Diagnostic Radiography Division

A patient cannot hold still and a limb image is blurred. Which exposure change is most appropriate?

Answer

Shorten the exposure time by using a higher mA (adjusting mAs or kVp to suit), and immobilise the limb. Do not simply repeat with the same time.

#291 · Diagnostic Radiography Division

Name three ways to control scatter.

Answer

Tight collimation, compression or an air gap, and a grid. Collimation is the first and cheapest tool.

#292 · Diagnostic Radiography Division

What is grid ratio?

Answer

The height of the lead strips divided by the gap between them. A higher ratio removes more scatter but needs more exposure and more accurate positioning.

#293 · Diagnostic Radiography Division

A non-grid technique uses 5 mAs. What mAs is needed with an 8:1 grid using a conversion factor of 4?

Answer

5 × 4 = 20 mAs. The dose rises fourfold in exchange for better contrast.

#294 · Diagnostic Radiography Division

When should a grid not be used?

Answer

On small paediatric parts and thin extremities, because the patient receives extra dose with little gain in image quality.

#295 · Diagnostic Radiography Division

An image has low density at both edges and normal density along the middle line. Which grid fault is likely?

Answer

An upside-down focused grid. Off-focus cutoff also lowers the edges, but upside-down gives very low density at both edges.

#296 · Diagnostic Radiography Division

An AEC image of a pelvis with a hip prosthesis over the centre chamber is overexposed around the metal. What is the best action?

Answer

Metal blocks the beam over the chamber so the exposure runs long. Select the side chambers or use manual factors rather than repeat the same exposure.

#297 · Diagnostic Radiography Division

What is the difference between CR and DR?

Answer

CR uses a photostimulable phosphor plate that a laser scans in a separate reader. DR uses a flat-panel detector that creates the digital image straight away.

#298 · Diagnostic Radiography Division

What does the exposure index tell you?

Answer

How much radiation reached the detector, not how much the patient received. It is compared with a target exposure index for the examination.

#299 · Diagnostic Radiography Division

What does a deviation index of 0, +1 and –1 mean?

Answer

0 is on target. +1 is about 26 percent more than the target. –1 is about 20 percent less than target. About ±1 is usually acceptable.

#300 · Diagnostic Radiography Division

A chest image looks excellent but the deviation index is +4. What should the radiographer do?

Answer

Treat it as overexposure even though the image looks good. Record it, review technique and AEC settings, and report through the local quality process.

#301 · Diagnostic Radiography Division

What is dose creep and how is it controlled?

Answer

A slow rise in exposures over time because digital images do not look too dark when overexposed. Check the exposure index on every image, audit indices and use technique charts.

#302 · Diagnostic Radiography Division

What causes quantum mottle and how is it reduced?

Answer

Too few photons reaching the detector, giving a grainy image. More mAs reduces it but raises dose, so repeat only if it affects the diagnosis.

#303 · Diagnostic Radiography Division

Can windowing recover detail from an underexposed digital image?

Answer

No. Windowing changes the displayed brightness and contrast only. It cannot add information the detector did not record or remove quantum mottle.

#304 · Diagnostic Radiography Division

The same white line appears in the same place on every image from one CR plate. What is the cause?

Answer

A scratch or dust on that plate. Clean it, and take the plate out of use if cleaning fails. Repeating on the same plate repeats the line.

#305 · Diagnostic Radiography Division

What does DAP measure and what unit is used?

Answer

Dose area product is the dose in air multiplied by beam area, in Gy·cm². It includes field size and is used for diagnostic reference levels.

#306 · Diagnostic Radiography Division

What is a diagnostic reference level, and is it a dose limit?

Answer

A benchmark for typical doses for a common examination on a standard-size patient. It is not a dose limit, and typical doses above it should be investigated and optimised.

#307 · Diagnostic Radiography Division

What Hounsfield unit value is given to water and to air in CT?

Answer

Water is 0 HU and air is about -1000 HU.

#308 · Diagnostic Radiography Division

In CT, what does DLP equal and what unit does it use?

Answer

DLP equals CTDIvol multiplied by scan length, in mGy cm.

#309 · Diagnostic Radiography Division

A scan has CTDIvol 10 mGy over 40 cm. What is the DLP?

Answer

DLP = 10 × 40 = 400 mGy cm.

#310 · Diagnostic Radiography Division

What happens to dose and noise when you double the mAs in CT?

Answer

Dose doubles and noise falls by about 30 percent. Noise falls with the square root of mAs.

#311 · Diagnostic Radiography Division

How much must mAs increase to halve image noise in CT?

Answer

Quadruple it, because noise falls with the square root of mAs.

#312 · Diagnostic Radiography Division

What does changing the CT window level and width alter?

Answer

Only the display of the image. It does not change the dose or the acquired data.

#313 · Diagnostic Radiography Division

A CT scan range was extended 10 cm beyond what the request needed. What is the effect on dose?

Answer

DLP rises because scan length rises, with no benefit. Scan only the range needed.

#314 · Diagnostic Radiography Division

Why must a department compare its median dose with the diagnostic reference level for an examination?

Answer

The comparison shows whether typical doses are higher than national practice. If so, technique is reviewed and optimised. It is a review tool, not a legal limit for one patient.

#315 · Diagnostic Radiography Division

A child needs a head CT and the image looks grainy. What is the best approach?

Answer

Use a size-based paediatric protocol and accept some noise if the clinical question can still be answered. Do not simply raise mAs.

#316 · Diagnostic Radiography Division

Which CT contrast phase is the usual routine phase for the abdomen?

Answer

The portal venous phase, roughly 60 to 70 seconds after injection.

#317 · Diagnostic Radiography Division

Which CT artefact looks like streaks around a hip prosthesis?

Answer

Metal artefact, with bright and dark streaks around the implant.

#318 · Diagnostic Radiography Division

Does a higher CT pitch raise or lower dose at a fixed tube current?

Answer

It lowers dose and covers the body faster, but can blur detail.

#319 · Diagnostic Radiography Division

What is the single most important MRI safety fact about the magnet?

Answer

The magnet is always on, even when no scan is running, so ferromagnetic objects can become projectiles.

#320 · Diagnostic Radiography Division

What do short TR and short TE produce in MRI?

Answer

A T1-weighted image, where fat is bright and simple fluid such as CSF is dark.

#321 · Diagnostic Radiography Division

Which MRI sequence suppresses CSF signal to show lesions near fluid spaces?

Answer

FLAIR (fluid attenuated inversion recovery).

#322 · Diagnostic Radiography Division

Which MRI sequence shows early stroke?

Answer

Diffusion weighted imaging (DWI), which shows restricted movement of water.

#323 · Diagnostic Radiography Division

What is Zone III in an MRI department?

Answer

A restricted zone where only screened staff and patients are allowed, because it leads to the scanner room.

#324 · Diagnostic Radiography Division

What does MR Conditional mean for an implant?

Answer

It is safe only under stated conditions, such as field strength and scan limits, which must be followed exactly.

#325 · Diagnostic Radiography Division

A patient says they have an old pacemaker and needs a knee MRI. What do you do first?

Answer

Stop, identify the exact device and leads, check their MR status, and involve the MRI safety lead, radiologist and cardiac team.

#326 · Diagnostic Radiography Division

What does SAR measure in MRI?

Answer

The rate at which the body absorbs radiofrequency energy as heat, in W/kg.

#327 · Diagnostic Radiography Division

Why is gadolinium used with caution in severe kidney failure?

Answer

Some older linear agents have been linked to nephrogenic systemic fibrosis. Current practice uses stable macrocyclic agents and checks kidney function.

#328 · Diagnostic Radiography Division

Does an iodine contrast allergy rule out gadolinium contrast?

Answer

No. Gadolinium is not an iodine contrast, but any previous gadolinium reaction must be recorded and communicated.

#329 · Diagnostic Radiography Division

In ultrasound, what trade-off comes with a higher probe frequency?

Answer

Better detail but less penetration, so it suits shallow structures.

#330 · Diagnostic Radiography Division

An ultrasound echo returns 130 µs after the pulse. What is the depth at 1540 m/s?

Answer

One-way time is 65 µs, so depth = 1540 × 65 × 10^-6 = 0.1001 m, about 10 cm.

#331 · Diagnostic Radiography Division

What artefact is a dark area behind a gallstone on ultrasound?

Answer

Acoustic shadowing.

#332 · Diagnostic Radiography Division

What artefact makes the area behind a simple cyst look brighter on ultrasound?

Answer

Posterior enhancement, because little sound is lost in the fluid.

#333 · Diagnostic Radiography Division

Which Doppler beam angle gives little or no shift, and what angle is advised for speed measurement?

Answer

At 90° to the vessel little or no shift is seen. Keep the angle at 60° or less for speed measurements.

#334 · Diagnostic Radiography Division

Why are mammography x-rays low kVp, and why is the breast compressed?

Answer

Low kVp gives high contrast between similar tissues. Compression spreads tissue, lowers dose, cuts scatter and reduces motion blur.

#335 · Diagnostic Radiography Division

What T-score defines osteoporosis on bone densitometry?

Answer

A T-score of -2.5 or below.

#336 · Nuclear Medicine Technology Division

Which decay mode of F-18 makes PET possible, and what does it produce?

Answer

Beta plus (positron) decay. The positron annihilates with an electron to give two 511 keV photons about 180 degrees apart.

#337 · Nuclear Medicine Technology Division

How does Tc-99m decay, and what is emitted?

Answer

By isomeric transition from a metastable state. It emits a 140 keV gamma ray with no particle, so the dose per detected photon is low.

#338 · Nuclear Medicine Technology Division

A department record shows a 10 mCi dose. What is this in MBq?

Answer

1 mCi is 37 MBq, so 10 mCi is 370 MBq.

#339 · Nuclear Medicine Technology Division

State the approximate physical half-lives of Tc-99m, F-18 and Mo-99.

Answer

Tc-99m about 6 hours, F-18 about 110 minutes, Mo-99 about 66 hours.

#340 · Nuclear Medicine Technology Division

A vial holds 1,200 MBq of Tc-99m at 08:00. What activity is present at 11:00?

Answer

Three hours is half a half-life. Factor 0.7071, so 1,200 × 0.7071 = 848.5 MBq.

#341 · Nuclear Medicine Technology Division

At 11:00 a vial has 848.5 MBq in 4 mL. What volume gives 600 MBq?

Answer

Concentration is 848.5 ÷ 4 = 212.1 MBq per mL. Volume is 600 ÷ 212.1 = 2.83 mL.

#342 · Nuclear Medicine Technology Division

A patient arrives late and the planned Tc-99m dose has decayed. What should the technologist do?

Answer

Recalculate the volume for the current time, measure the syringe in the dose calibrator, and record the activity and time. Do not give the original volume.

#343 · Nuclear Medicine Technology Division

Write the formula linking effective, physical and biological half-life.

Answer

1 ÷ Te = 1 ÷ Tp + 1 ÷ Tb, or Te = (Tp × Tb) ÷ (Tp + Tb).

#344 · Nuclear Medicine Technology Division

A tracer has Tp = 6 hours and Tb = 6 hours. What is the effective half-life?

Answer

Te = (6 × 6) ÷ (6 + 6) = 3 hours.

#345 · Nuclear Medicine Technology Division

A calculation gives an effective half-life longer than the physical half-life. What does this tell you?

Answer

The sum is wrong. The effective half-life is always shorter than both the physical and the biological half-life.

#346 · Nuclear Medicine Technology Division

How does the Mo-99/Tc-99m generator separate Tc-99m?

Answer

Mo-99 is bound to an alumina column. Saline drawn through by an evacuated vial washes off Tc-99m as sodium pertechnetate.

#347 · Nuclear Medicine Technology Division

What type of equilibrium exists in the Mo-99/Tc-99m generator, and when does Tc-99m peak?

Answer

Transient equilibrium, because the parent half-life is about 11 times the daughter's. Tc-99m peaks about 23 hours after elution.

#348 · Nuclear Medicine Technology Division

How is Mo-99 breakthrough measured in the eluate?

Answer

The first eluate vial is placed in a lead sleeve in the dose calibrator. The sleeve blocks Tc-99m 140 keV photons but passes the higher energy Mo-99 photons.

#349 · Nuclear Medicine Technology Division

Why is aluminium breakthrough in the eluate a problem?

Answer

Aluminium ions interfere with labelling and can clump some kits, such as sulphur colloid and MAA. It is checked with a colorimetric spot test.

#350 · Nuclear Medicine Technology Division

A labelled kit fails radiochemical purity. What is the correct action?

Answer

Do not give the dose. Investigate the cause, such as an old eluate or aluminium breakthrough, and inform the radiopharmacist or physician.

#351 · Nuclear Medicine Technology Division

What does a collimator do in a gamma camera?

Answer

It is a lead plate with holes that passes only photons travelling along the hole direction, so the crystal forms an image.

#352 · Nuclear Medicine Technology Division

Which collimator gives better spatial resolution, LEHR or LEGP, and what is the cost?

Answer

LEHR gives better resolution but lower sensitivity, because fewer photons pass through its narrower or longer holes.

#353 · Nuclear Medicine Technology Division

What does the NaI(Tl) crystal do, and what does a crack in it cause?

Answer

It turns absorbed gamma rays into flashes of light. A crack causes a persistent defect, often a cold area, on the image.

#354 · Nuclear Medicine Technology Division

What is the role of the photomultiplier tubes?

Answer

They turn the light flash into electrical signals. Comparing neighbouring tubes gives the X and Y position, and the summed signal gives the photon energy.

#355 · Nuclear Medicine Technology Division

What energy range does a 20 percent window on Tc-99m cover?

Answer

140 ± 14 keV, which is 126 to 154 keV. It rejects most scattered photons.

#356 · Nuclear Medicine Technology Division

Describe true, scatter and random coincidences in PET.

Answer

True: both photons from one annihilation, unscattered. Scatter: a photon changed direction, giving a wrong line. Random: photons from different annihilations detected within the timing window.

#357 · Nuclear Medicine Technology Division

Why does PET not need a collimator?

Answer

The scanner uses coincidence detection of two 511 keV photons in a short timing window, which defines the line of response electronically.

#358 · Nuclear Medicine Technology Division

Calculate the SUV: tissue 20 kBq/mL, decay-corrected dose 370 MBq, weight 74 kg.

Answer

Dose per gram is 370,000 ÷ 74,000 = 5 kBq/g. SUV = 20 ÷ 5 = 4.0.

#359 · Nuclear Medicine Technology Division

A hot spot on an attenuation-corrected PET image lies next to a metal implant. What is the best next step?

Answer

Check the non-attenuation-corrected images. Metal causes a CT artefact that can create a false hot spot after correction.

#360 · Nuclear Medicine Technology Division

Daily uniformity flood on a gamma camera shows a new non-uniform area. What should the practitioner do?

Answer

Not start patient studies on that detector. Repeat the test, check the energy peak, and if it fails again, take the camera out of use and report it.

#361 · Nuclear Medicine Technology Division

What does a centre of rotation error cause on SPECT, and how is it detected?

Answer

Blurred or ring-shaped artefacts in reconstructed slices. It is detected by a COR test with a point source.

#362 · Nuclear Medicine Technology Division

What does the dose calibrator constancy test use, and how often is it done?

Answer

A long half-life reference source such as Cs-137, measured daily. The reading should match the expected decayed value.

#363 · Nuclear Medicine Technology Division

A count of 10,000 is recorded in a well counter. What is the standard deviation and percentage error?

Answer

SD = √10,000 = 100, so the error is 100 ÷ 10,000 = 1 percent.

#364 · Nuclear Medicine Technology Division

What is the half-life of technetium-99m, and what gamma energy does it emit?

Answer

About 6 hours, with a 140 keV gamma ray, which suits a gamma camera.

#365 · Nuclear Medicine Technology Division

In a Mo-99/Tc-99m generator, what happens when sterile saline is passed through the column?

Answer

Tc-99m comes off the column as sodium pertechnetate. This step is called an elution.

#366 · Nuclear Medicine Technology Division

What is Mo-99 breakthrough, and what is a common limit?

Answer

It is Mo-99 leaking into the eluate. A common limit is 0.15 kBq of Mo-99 per MBq of Tc-99m, as set in the product information.

#367 · Nuclear Medicine Technology Division

Why is aluminium breakthrough in the eluate a problem, and what is a common limit?

Answer

Aluminium can clump agents such as sulfur colloid and MAA. A common limit is 10 micrograms per mL.

#368 · Nuclear Medicine Technology Division

What does a cold kit contain besides the carrier molecule?

Answer

A reducing agent, usually stannous chloride, and a buffer. The pertechnetate is reduced so it can bind to the carrier.

#369 · Nuclear Medicine Technology Division

A busy technologist wants to skip wiping the vial septum to reduce time near the source. What is the right decision?

Answer

Do not skip it. Wipe the septum with a 70 percent alcohol swab and let it dry. Reduce dose with shielding, distance and planning, not by cutting aseptic steps.

#370 · Nuclear Medicine Technology Division

Define radiochemical purity.

Answer

The percentage of the radioactivity that is in the intended chemical form. The usual target is about 90 to 95 percent or higher, as set by the kit information.

#371 · Nuclear Medicine Technology Division

Name the two common impurities in a Tc-99m kit.

Answer

Free pertechnetate and reduced hydrolysed technetium (Tc-99m colloid).

#372 · Nuclear Medicine Technology Division

Counts on an ITLC strip are 3,000 at the top piece and 57,000 at the bottom piece. What percentage is free pertechnetate in saline?

Answer

3,000 ÷ 60,000 = 0.05, so 5 percent free pertechnetate runs with the solvent front.

#373 · Nuclear Medicine Technology Division

A kit fails its radiochemical purity test. What should you do?

Answer

Do not use it on a patient. Quarantine the vial, record it and follow the department procedure to find the cause.

#374 · Nuclear Medicine Technology Division

A lung scan shows a bright thyroid and stomach. What problem should you suspect first?

Answer

Free pertechnetate from a labelling or stability problem, not lung disease. Check the radiopharmaceutical before blaming the camera.

#375 · Nuclear Medicine Technology Division

How does Tc-99m MDP localise in the body?

Answer

It binds (chemisorbs) to hydroxyapatite where bone is being laid down and where blood flow is high.

#376 · Nuclear Medicine Technology Division

Why does Tc-99m MAA show lung perfusion?

Answer

MAA particles of about 10 to 90 micrometres lodge in lung capillaries after intravenous injection.

#377 · Nuclear Medicine Technology Division

In which patients is the MAA particle count reduced?

Answer

Children, people with pulmonary hypertension and people with a right to left shunt.

#378 · Nuclear Medicine Technology Division

What is the difference in how Tc-99m DTPA and Tc-99m MAG3 are cleared by the kidney?

Answer

DTPA is cleared by glomerular filtration. MAG3 is cleared mainly by tubular secretion.

#379 · Nuclear Medicine Technology Division

What does Tc-99m DMSA show, and why?

Answer

The renal cortex. It binds to proximal tubule cells and stays there, so late images show scarring or functioning cortex.

#380 · Nuclear Medicine Technology Division

Why do sestamibi and tetrofosmin show heart muscle perfusion?

Answer

They are lipophilic cations that enter heart muscle cells in proportion to blood flow and are held in mitochondria.

#381 · Nuclear Medicine Technology Division

Why does F-18 FDG light up many cancers?

Answer

It is a glucose analogue. It is phosphorylated and trapped in cells that use a lot of glucose. F-18 has a half-life of about 110 minutes.

#382 · Nuclear Medicine Technology Division

A vial holds 600 MBq of Tc-99m at 08:00. What is the activity at 14:00?

Answer

6 hours is one half-life, so 600 × 1/2 = 300 MBq.

#383 · Nuclear Medicine Technology Division

State the decay equation using the half-life.

Answer

A = A0 × (1/2)^(t ÷ T½), with t and T½ in the same unit.

#384 · Nuclear Medicine Technology Division

A vial has 932 MBq in 4.0 mL at injection time. How much volume gives 740 MBq?

Answer

932 ÷ 4.0 = 233 MBq per mL. 740 ÷ 233 = 3.18 mL, about 3.2 mL.

#385 · Nuclear Medicine Technology Division

A child weighs 20 kg and the adult activity is 600 MBq for 70 kg. What is the weight scaled activity, and what else must you check?

Answer

600 × 20 ÷ 70 = 171 MBq. Check the minimum activity in the paediatric dose card or department protocol and give the minimum if it is higher.

#386 · Nuclear Medicine Technology Division

A patient with diabetes has blood glucose above the department limit before FDG PET. What is the best action?

Answer

Do not inject. Follow protocol to reschedule or correct glucose. Injecting anyway risks a non-diagnostic scan.

#387 · Nuclear Medicine Technology Division

Why must caffeine be withheld before adenosine or regadenoson stress imaging?

Answer

Caffeine blocks adenosine receptors and can make the stress test fail. Withhold for the time in the protocol.

#388 · Nuclear Medicine Technology Division

Why can opioid pain medicine affect a mebrofenin hepatobiliary scan?

Answer

It can cause spasm of the sphincter of Oddi, so the bile duct looks blocked. Review it before the scan.

#389 · Nuclear Medicine Technology Division

List the checks before giving any radiopharmaceutical.

Answer

Right patient (three identifiers), agent, activity measured in a dose calibrator, route and time. Also record pregnancy, breastfeeding and allergy status, then measure the residual after injection.

#390 · Nuclear Medicine Technology Division

You suspect extravasation during a diagnostic injection. What is the first action?

Answer

Stop the injection, record the site and estimated amount, tell the nuclear medicine physician, and consider imaging the site. Do not repeat a full dose without the physician's decision.

#391 · Nuclear Medicine Technology Division

What is the best advice for breastfeeding after I-131 therapy?

Answer

Breastfeeding must stop for that infant, and therapy is not given to a woman who is still breastfeeding.

#392 · Nuclear Medicine Technology Division

What is a radiopharmaceutical?

Answer

A medicine that carries a radioactive label (radionuclide) to a particular organ or process so it can be imaged or treated.

#393 · Nuclear Medicine Technology Division

How many patient identifiers should be used before giving a radiopharmaceutical?

Answer

At least three, for example full name, date of birth and medical record number, matched to the request and the syringe label.

#394 · Nuclear Medicine Technology Division

What is the physical half life of Tc-99m?

Answer

About 6 hours.

#395 · Nuclear Medicine Technology Division

What is the physical half life of F-18?

Answer

About 110 minutes.

#396 · Nuclear Medicine Technology Division

A syringe holds 800 MBq of Tc-99m at 0800. How much is left at 2000?

Answer

200 MBq, because 12 hours is two half lives: 800 to 400 to 200.

#397 · Nuclear Medicine Technology Division

A syringe holds 400 MBq of F-18 at 0900. How much remains at 1050?

Answer

About 200 MBq, because 110 minutes is one half life.

#398 · Nuclear Medicine Technology Division

Which radiopharmaceutical is used for a standard bone scan?

Answer

Tc-99m methylene diphosphonate (MDP), taken up where bone is being laid down.

#399 · Nuclear Medicine Technology Division

Why is a patient told to drink extra fluid after a bone scan injection?

Answer

It clears unbound tracer and lowers the bladder dose, and frequent voiding helps further.

#400 · Nuclear Medicine Technology Division

A bone scan shows a hot spot on the skin. What is the likely cause and best action?

Answer

Urine contamination. Wash the area, remove contaminated clothing and repeat the view.

#401 · Nuclear Medicine Technology Division

What does thyroid uptake of Tc-99m pertechnetate depend on?

Answer

The thyroid trapping the tracer, which recent iodinated contrast, some thyroid drugs and amiodarone can reduce.

#402 · Nuclear Medicine Technology Division

A patient due for thyroid imaging had iodinated contrast last week. What should you do?

Answer

Tell the physician before injecting and follow their decision, because contrast can lower uptake.

#403 · Nuclear Medicine Technology Division

What does a myocardial perfusion study compare?

Answer

Blood flow to the heart muscle at rest and under stress, to separate ischaemia from scar.

#404 · Nuclear Medicine Technology Division

Which tracers are usually used for myocardial perfusion imaging?

Answer

Tc-99m sestamibi or tetrofosmin.

#405 · Nuclear Medicine Technology Division

When is pharmacological stress used instead of exercise?

Answer

When the patient cannot exercise to an adequate heart rate, or has a left bundle branch block or paced rhythm.

#406 · Nuclear Medicine Technology Division

Name two contraindications to adenosine or regadenoson.

Answer

Active bronchospasm (asthma with wheeze or severe COPD) and high degree AV block without a pacemaker.

#407 · Nuclear Medicine Technology Division

Why must caffeine be withheld before adenosine stress?

Answer

Caffeine blocks the receptor adenosine acts on, so the stress may be falsely normal. Hold it for the department's period, commonly 12 to 24 hours.

#408 · Nuclear Medicine Technology Division

A patient due for adenosine stress had two coffees this morning. What is the best action?

Answer

Stop, inform the nuclear medicine physician and reschedule. Do not give a larger dose.

#409 · Nuclear Medicine Technology Division

Which stress agent is usually the alternative when a vasodilator is contraindicated?

Answer

Dobutamine, which raises heart rate and contractility.

#410 · Nuclear Medicine Technology Division

During a stress test the patient has chest pain with ECG change. What should you do?

Answer

Stop the stress, call for help and start the emergency response with the clinician present.

#411 · Nuclear Medicine Technology Division

In a V/Q scan, what does a ventilated but not perfused segment suggest?

Answer

A mismatch, which raises suspicion of pulmonary embolism.

#412 · Nuclear Medicine Technology Division

Which agent is used for the perfusion part of a lung V/Q scan?

Answer

Tc-99m macroaggregated albumin (MAA), which lodges in small lung vessels in proportion to blood flow.

#413 · Nuclear Medicine Technology Division

What does DMSA show in a kidney study?

Answer

Cortical scarring and differential function, as it binds to the kidney cortex.

#414 · Nuclear Medicine Technology Division

Why is a diuretic given during a MAG3 study?

Answer

To test whether a full, enlarged collecting system is obstructed or just baggy. It is given when the system is full.

#415 · Nuclear Medicine Technology Division

Why must a patient fast before a hepatobiliary scan?

Answer

A recent meal contracts the gallbladder and stops it filling, which can mimic disease.

#416 · Nuclear Medicine Technology Division

What labels the tracer in a gastrointestinal bleed study?

Answer

The patient's own red blood cells labelled with Tc-99m, so the tracer stays in the blood until it leaks into the bowel.

#417 · Nuclear Medicine Technology Division

What preparation does FDG PET/CT require?

Answer

Fasting for about 4 to 6 hours with water allowed, a measured blood glucose, and about 60 minutes of quiet warm rest after injection.

#418 · Nuclear Medicine Technology Division

Why is the patient kept warm and still during FDG uptake time?

Answer

Warmth lowers brown fat uptake, and rest avoids muscle uptake that could mimic disease.

#419 · Nuclear Medicine Technology Division

A diabetic patient's blood glucose is above the department limit before FDG. What is the best action?

Answer

Delay and tell the physician, who sets the plan. Do not change insulin or give it on your own authority.

#420 · Nuclear Medicine Technology Division

What should written release advice after I-131 therapy include?

Answer

Keep distance from others (especially children and pregnant women), sleep alone, hygiene and toilet rules, avoid public transport, stay off work for the stated days, and carry a treatment card.

#421 · Radiation Therapy Division

What is the gray (Gy), and how does it relate to centigray?

Answer

One gray is 1 joule of energy absorbed per kilogram of tissue. 1 Gy equals 100 cGy.

#422 · Radiation Therapy Division

What does a monitor unit (MU) on a linear accelerator measure?

Answer

It is a reading from the ionisation chambers in the machine head. The machine is calibrated so a set number of MU gives a known dose under reference conditions.

#423 · Radiation Therapy Division

List the main linac parts in beam order from the electron gun to the collimators.

Answer

Electron gun, accelerating waveguide, bending magnet, target, flattening filter, ionisation chambers, then jaws and multileaf collimator.

#424 · Radiation Therapy Division

What is the isocentre, and how far is it usually from the source?

Answer

It is the point where the gantry, collimator and couch axes meet. It is usually 100 cm from the source.

#425 · Radiation Therapy Division

How is a photon beam made inside a linac?

Answer

Accelerated electrons strike a heavy metal target such as tungsten and make X-rays by bremsstrahlung.

#426 · Radiation Therapy Division

What changes in the linac head to produce an electron beam instead of photons?

Answer

The target and flattening filter are moved out and a scattering foil and an applicator with a shaped cutout are used.

#427 · Radiation Therapy Division

What is dmax, and roughly where is it for 6 MV and 15 MV photons?

Answer

It is the depth of maximum dose. It is about 1.5 cm for 6 MV and about 3 cm for 15 MV.

#428 · Radiation Therapy Division

Why do megavoltage photon beams spare the skin?

Answer

The dose builds up over the first centimetres as forward moving electrons deposit energy, so the surface dose is low.

#429 · Radiation Therapy Division

A skin lesion only 1 cm deep is planned with a 15 MV photon beam. What is the problem?

Answer

The build-up region is deeper than the target, so the target receives too little dose. A lower energy photon beam, an electron beam or bolus is more suitable.

#430 · Radiation Therapy Division

What is the rough rule for the practical range of an electron beam in tissue?

Answer

The practical range in centimetres is about the energy in MeV divided by 2. The 90 percent depth is about the energy divided by 3 or 4.

#431 · Radiation Therapy Division

Why are electron beams chosen for superficial targets?

Answer

They keep a high dose to a set depth and then fall steeply, so little dose reaches deeper tissue beyond the target.

#432 · Radiation Therapy Division

How do superficial and orthovoltage X-ray units differ from a megavoltage linac in dose pattern?

Answer

They give kilovoltage X-rays with the highest dose at the surface, no build-up and no skin sparing. Bone absorbs more dose than soft tissue.

#433 · Radiation Therapy Division

What does percentage depth dose (PDD) describe?

Answer

It is the dose at a depth as a percentage of the dose at dmax on the central axis for a stated energy, field size and SSD.

#434 · Radiation Therapy Division

What happens to PDD at a given depth as photon energy increases?

Answer

It increases, because the higher energy beam is more penetrating.

#435 · Radiation Therapy Division

State the inverse square law for dose rate from a point source.

Answer

Dose rate changes in proportion to 1 divided by the distance squared.

#436 · Radiation Therapy Division

The dose rate is 1.00 at 100 cm. What is it at 110 cm?

Answer

It is $(100 \div 110)^2$, which is 10,000 ÷ 12,100 = 0.826, about 17 percent lower.

#437 · Radiation Therapy Division

A patient is set up 10 cm too far from the source and the MU are unchanged. What happens?

Answer

The patient is under-treated, because the dose rate is lower at the greater distance.

#438 · Radiation Therapy Division

What does tissue phantom ratio (TPR) compare, and when is it used?

Answer

It compares the dose at a depth with the dose at the same point at a reference depth, independent of SSD. It suits isocentric techniques.

#439 · Radiation Therapy Division

A single field has PDD 0.67 at the prescription depth and a field size factor of 1.02. Dose is 200 cGy at 1 cGy per MU at dmax. How many MU?

Answer

200 ÷ 0.67 = 298.5, then ÷ 1.02 = 292.6, so 293 MU. Check: 293 × 1.02 × 0.67 = 200.2 cGy.

#440 · Radiation Therapy Division

What is the purpose of bolus, and what must be checked when it is used?

Answer

It removes skin sparing and fills irregular contours. It must sit flat against the skin with no air gap.

#441 · Radiation Therapy Division

The therapist finds an air gap under bolus before treatment. What is the best action?

Answer

Refit the bolus flat against the skin, and tell the team if the gap cannot be removed. An air gap lowers the dose near the skin.

#442 · Radiation Therapy Division

What do the jaws and the multileaf collimator each do?

Answer

The jaws set the maximum rectangular field and the MLC leaves shape the beam to the target.

#443 · Radiation Therapy Division

Which source is common for HDR brachytherapy, and what is its half-life?

Answer

Iridium-192, with a half-life of about 74 days, so it is replaced several times a year.

#444 · Radiation Therapy Division

During HDR brachytherapy the source does not return to the safe. What is the first response?

Answer

Do not enter the room. Use the emergency retract or stop, check the room monitor, and call the radiation safety officer and physicist.

#445 · Radiation Therapy Division

Which imaging on treatment shows soft tissue best: EPID portal image, kV planar image or CBCT?

Answer

CBCT shows soft tissue best. The portal image and kV planar image show bone better.

#446 · Radiation Therapy Division

A daily output or interlock check fails before the first patient. What should the therapist do?

Answer

Do not treat. Report and record the failure and wait for the physicist to clear the machine.

#447 · Radiation Therapy Division

Why must a door interlock never be bypassed to save time?

Answer

It prevents the beam starting when the door is open, so bypassing it risks staff and public exposure. If it cannot be cleared, stop and call the physicist.

#448 · Radiation Therapy Division

What does the gross tumour volume (GTV) describe in radiotherapy planning?

Answer

The tumour that can be seen or felt on imaging or examination.

#449 · Radiation Therapy Division

What does the clinical target volume (CTV) add to the GTV?

Answer

A margin for microscopic spread of disease that cannot be seen but is likely to be present.

#450 · Radiation Therapy Division

What is the planning target volume (PTV) and why is it used?

Answer

The CTV (or ITV) plus a margin for set-up error and uncertainty. It makes sure the CTV truly receives the dose on every day of treatment.

#451 · Radiation Therapy Division

What is an internal target volume (ITV)?

Answer

The CTV plus an allowance for movement of the target inside the body, for example with breathing.

#452 · Radiation Therapy Division

Why is a planning organ at risk volume (PRV) used for the spinal cord?

Answer

The cord is a serial organ, so a small overdose could cause serious harm. The PRV adds a margin around it to allow for uncertainty.

#453 · Radiation Therapy Division

What is the difference between a systematic and a random set-up error?

Answer

A systematic error is the same every day and shifts the whole course in one direction. A random error varies day to day and blurs the dose. Systematic errors are more dangerous.

#454 · Radiation Therapy Division

A patient's thermoplastic mask no longer fits after weight loss. What is the best action before treating?

Answer

Stop, tell the treating team and arrange a new mask or repeat simulation if needed. Do not force the mask or treat anyway.

#455 · Radiation Therapy Division

What does intensity-modulated radiation therapy (IMRT) do differently from 3D conformal therapy?

Answer

It varies the intensity across each beam by moving the multileaf collimator during delivery, so dose can wrap around concave shapes and spare nearby organs.

#456 · Radiation Therapy Division

What is volumetric-modulated arc therapy (VMAT)?

Answer

IMRT delivered while the gantry rotates in one or more arcs. It is usually faster than fixed-field IMRT.

#457 · Radiation Therapy Division

Why are set-up checks and image guidance stricter for SBRT and SRS?

Answer

Very high doses are given in one to a few fractions, so any error forms a large part of the whole treatment.

#458 · Radiation Therapy Division

What does a dose-volume histogram (DVH) show?

Answer

The percentage of a structure's volume that receives at least a given dose, plotted for each structure.

#459 · Radiation Therapy Division

How should a good target curve and a good organ at risk curve look on a DVH?

Answer

The target curve stays high and then drops sharply near the prescription dose. The organ at risk curve falls away quickly, so little of the organ gets a high dose.

#460 · Radiation Therapy Division

Which dose matters most for a serial organ such as the spinal cord, and which for a parallel organ such as lung?

Answer

For a serial organ the maximum point dose matters most. For a parallel organ the volume receiving a moderate dose matters most.

#461 · Radiation Therapy Division

A plan covers the PTV well but breaks a spinal cord constraint. What should the practitioner do?

Answer

Flag it to the planner and the radiation oncologist so a decision is made. Do not accept the plan quietly or trade the cord limit away without the oncologist deciding.

#462 · Radiation Therapy Division

List the items a complete radiotherapy prescription must include.

Answer

Site, total dose, dose per fraction, number of fractions, technique, energy, and the dose specification point or volume.

#463 · Radiation Therapy Division

What is conventional fractionation?

Answer

About 1.8 to 2 Gy per fraction, one fraction a day, five days a week, over several weeks.

#464 · Radiation Therapy Division

Why is the total dose split into fractions?

Answer

Between treatments healthy tissue repairs itself better than tumour does, so splitting the dose spares normal tissue.

#465 · Radiation Therapy Division

What is the formula for biologically effective dose (BED)?

Answer

BED = n × d × (1 + d ÷ (alpha/beta)), where n is the number of fractions and d is the dose per fraction in Gy.

#466 · Radiation Therapy Division

Calculate the BED for 30 fractions of 2 Gy with an alpha/beta of 10 Gy.

Answer

Total 60 Gy. d ÷ (alpha/beta) = 0.2. BED = 60 × 1.2 = 72 Gy.

#467 · Radiation Therapy Division

Why do late-reacting tissues such as spinal cord care more about fraction size?

Answer

They have a low alpha/beta ratio, so larger fractions raise late damage faster than tumour effect, and the total dose must be reduced.

#468 · Radiation Therapy Division

Before any beam is delivered, how should patient identity be confirmed?

Answer

Use at least three identifiers, such as full name, date of birth and record number, asked of the patient where they can answer.

#469 · Radiation Therapy Division

What is the purpose of a time-out before treatment?

Answer

A short pause where the team confirms aloud the patient, site, technique and special instructions such as bolus or a cardiac device.

#470 · Radiation Therapy Division

The record-and-verify system shows a warning. What is the best action?

Answer

Find out why before continuing. Never override a warning you do not understand.

#471 · Radiation Therapy Division

A CBCT shows a 9 mm shift when the department's action level is 5 mm. What should the practitioner do?

Answer

Stop and check the set-up, patient position and image match, and involve senior staff or the oncologist as the protocol says. Do not simply apply the shift.

#472 · Radiation Therapy Division

A prescription is 66 Gy in 33 fractions. After 14 fractions, how much dose remains?

Answer

Dose per fraction is 2 Gy, delivered is 28 Gy, so 38 Gy remains, which is 19 fractions.

#473 · Radiation Therapy Division

A patient misses a treatment day. Who decides how the dose is made up?

Answer

The radiation oncologist. The radiation therapist records the missed fraction and refers the decision.

#474 · Radiation Therapy Division

A patient with a pacemaker arrives and no one knew about the device. What happens next?

Answer

Treatment waits until the radiation oncologist has reviewed the plan, with a dose limit set by local protocol and advice from the cardiology or device clinic.

#475 · Radiation Therapy Division

What does the N in TNM staging describe?

Answer

Whether cancer cells are present in regional lymph nodes.

#476 · Radiation Therapy Division

What does the M in TNM staging describe?

Answer

Whether there are distant metastases, meaning cancer deposits in organs away from the primary tumour.

#477 · Radiation Therapy Division

What is the aim of curative (radical) radiation therapy and what does it accept?

Answer

The aim is to cure the cancer. It uses higher doses over several weeks and accepts more side effects.

#478 · Radiation Therapy Division

What is the aim of palliative radiation therapy?

Answer

To relieve symptoms such as pain or bleeding, using lower doses and shorter courses with few side effects.

#479 · Radiation Therapy Division

A patient on palliative treatment for bone pain asks if it will cure the cancer. What is the best response?

Answer

Be honest and kind: the aim is to ease pain, it may take a couple of weeks to work, and the oncology team can discuss the wider plan.

#480 · Radiation Therapy Division

Name the four Rs of radiobiology.

Answer

Repair, repopulation, reassortment (redistribution) and reoxygenation.

#481 · Radiation Therapy Division

What is the therapeutic ratio?

Answer

The gap between the dose that controls the tumour and the dose that causes unacceptable normal tissue damage. Radiation therapy tries to widen it.

#482 · Radiation Therapy Division

Why are hypoxic tumour cells a problem for radiation therapy?

Answer

They are roughly two and a half to three times harder to kill with X-rays than well oxygenated cells.

#483 · Radiation Therapy Division

Which type of DNA damage is most likely to kill a cell?

Answer

A double strand break, because it is much harder to repair than a single strand break.

#484 · Radiation Therapy Division

Why is the total dose divided into daily fractions?

Answer

It lets normal tissue repair between fractions and exposes tumour cells to reoxygenation and reassortment, sparing late responding tissue.

#485 · Radiation Therapy Division

A prescription is 60 Gy in 30 fractions and 12 fractions are done. How much dose remains?

Answer

Dose per fraction is 2 Gy. Delivered is 24 Gy, so 36 Gy remains over 18 fractions.

#486 · Radiation Therapy Division

In the BED formula, which two quantities decide how much a larger dose per fraction raises the biological effect?

Answer

The dose per fraction d and the α/β ratio of the tissue. A bigger d, or a smaller α/β ratio, makes the term $d \div (\alpha/\beta)$ larger, so BED rises more.

#487 · Radiation Therapy Division

What α/β values are typically used for tumours and for late responding tissue?

Answer

About 10 Gy for tumours and early responding tissue, and about 3 Gy for late responding tissue.

#488 · Radiation Therapy Division

What is the BED to late tissue (α/β 3 Gy) for 60 Gy given as 20 fractions of 3 Gy?

Answer

60 × (1 + 3 ÷ 3) = 60 × 2 = 120 Gy.

#489 · Radiation Therapy Division

Why must a hypofractionated schedule use a lower total dose than a conventional one?

Answer

Larger fractions are biologically stronger, and they raise the late tissue effect more than the tumour effect.

#490 · Radiation Therapy Division

A patient misses several treatment days. What should the therapist do?

Answer

Report the missed fractions to the radiation oncologist, who decides how to compensate, because tumour cells repopulate during gaps.

#491 · Radiation Therapy Division

Which tissues show acute radiation reactions, and when do they appear?

Answer

Fast dividing tissues such as skin, mucosa and bowel lining, during treatment or within about three months.

#492 · Radiation Therapy Division

What is mucositis and which patients get it most?

Answer

Soreness and ulcers in the mouth and throat, most common in head and neck patients.

#493 · Radiation Therapy Division

A head and neck patient in week 4 has painful swallowing and weight loss. What is the best action?

Answer

Report to the nurse or radiation oncologist the same day and request dietitian review. Do not just say it will settle.

#494 · Radiation Therapy Division

Give two pieces of skin care advice during radiation therapy.

Answer

Wash gently with lukewarm water and mild soap and pat dry, and avoid sun, hot packs, tight clothing and adhesives on the area.

#495 · Radiation Therapy Division

Why can radiation reactions be stronger when chemotherapy is given at the same time?

Answer

Concurrent chemotherapy can strengthen radiation reactions, so the patient needs close monitoring and prompt reporting of symptoms.

#496 · Radiation Therapy Division

Why are late effects often permanent?

Answer

They occur in slowly dividing tissues such as spinal cord, kidney and connective tissue, months to years after treatment.

#497 · Radiation Therapy Division

Name the warning signs of spinal cord compression.

Answer

New or worsening back pain, band-like pain, leg weakness or heaviness, numbness, difficulty walking, or new bladder or bowel change.

#498 · Radiation Therapy Division

A lung cancer patient says their legs feel heavy and strange since yesterday. What is the best action?

Answer

Stop, keep the patient safe, and report to the radiation oncologist immediately, following the local procedure.

#499 · Radiation Therapy Division

What signs suggest superior vena cava obstruction?

Answer

Swelling of the face, neck and arms, breathlessness, distended neck veins, and sometimes headache.

#500 · Radiation Therapy Division

What three principles protect staff around brachytherapy sources?

Answer

Time, distance and shielding, with source accounting and a rehearsed emergency procedure for a source that does not retract.